Test Code Masters

November 20, 2025

[96]:
```
import subprocess
import os
import numpy as np
from astropy.io import fits
import matplotlib.pyplot as plt
import shutil
from astroquery.heasarc import Heasarc
from astropy.coordinates import SkyCoord
from astropy.time import Time
from astropy.table import Table
from pathlib import Path
from datetime import datetime
import pandas as pd
from datetime import timedelta
```

[2]:
```
star_name = "1RXS_J154439.4-112820"
ObsID_wanted = 2




os.chdir("/home/cananden/")
star_folder = f"{star_name}"

filter_min = 30   # eV
filter_max = 1200 # eV



os.chdir("nicer-data")

if os.path.isdir(f"{star_folder}"):
    None
else:
    !mkdir {star_folder}


os.chdir(f"{star_folder}")
```

```python
print(f"Inside {star_folder}")


pos = SkyCoord.from_name(f"{star_name}")
RA = pos.ra.deg
Dec = pos.dec.deg
tab = Heasarc.query_region(pos, catalog="nicermastr", radius="3 arcmin")
tab = tab[tab["time"] > 0]
tab.sort("time")
tab["name", "obsid", "ra", "dec"][:].pprint()

print("")
print(f"RA:{RA}, Dec {Dec}")
print("")
heasarc = Heasarc()


table = heasarc.query_region(pos, catalog="nicermastr", radius="3 arcmin")

table = table[table["time"] > 0]

table.sort("time")
table = table[table["name"] == star_name]

Epoch_array = []
for mjd in table["time"]:
    t = Time(mjd, format="mjd")
    Epoch_array.append(t.isot)

Epoch_array = np.array(Epoch_array)

ObsID_array = np.array(table["obsid"])
ObsID_date = []
for mjd in table["public_date"]:
    t = Time(mjd, format="mjd")
    ObsID_date.append(f"{t.datetime.year}_{t.datetime.month:02d}")

ObsID_date = np.array(ObsID_date)


Observation_date = []
for mjd in table["processing_date"]:
    t = Time(mjd, format="mjd")
    Observation_date.append(f"{t.datetime.year}_{t.datetime.month:02d}")


Observation_array = np.array(Observation_date)
```

```
ObsID_complete = Table(
    [ObsID_array, ObsID_date, Observation_array, Epoch_array],
    names=("ObsID", "Public Date", "Processing Date", "Epoch")
)
print(ObsID_complete)

ObsID_run = min(ObsID_wanted, len(ObsID_array))
links = Heasarc.locate_data(table)

seen = set()
downloaded = 0

for row in links:
    if downloaded >= ObsID_wanted:
        break

    url = row["access_url"]
    if url in seen:
        continue
    seen.add(url)

    parts = url.rstrip("/").split("/")
    obsid = parts[-1]
    folder = os.path.join(parts[-2], obsid)
    if os.path.isdir(folder):
        continue

    !~/download_wget.pl {url}

    downloaded += 1
```

```
Inside 1RXS_J154439.4-112820
         name           obsid        ra       dec
                                     deg       deg
--------------------- ---------- --------- ---------
1RXS_J154439.4-112820 0030120101 236.14130 -11.44195
1RXS_J154439.4-112820 0030120102 236.19660 -11.46890
1RXS_J154439.4-112820 0030120103 236.16400 -11.46540
1RXS_J154439.4-112820 0030120104 236.16410 -11.46543
1RXS_J154439.4-112820 0030120105 236.16380 -11.46548
1RXS_J154439.4-112820 0030120106 236.16450 -11.46568
1RXS_J154439.4-112820 0030120107 236.16420 -11.46599
1RXS_J154439.4-112820 0030120108 236.16430 -11.46543
1RXS_J154439.4-112820 0030120109 236.16430 -11.46547
                   …          …        …        …
1RXS_J154439.4-112820 6030120101 236.16400 -11.46509
1RXS_J154439.4-112820 6030120102 236.16420 -11.46506
```

```
1RXS_J154439.4-112820 6030120103 236.16370 -11.46467
1RXS_J154439.4-112820 7030120101 236.16440 -11.46484
1RXS_J154439.4-112820 7030120102 236.16440 -11.46489
    3FGL_J1544.6-1125 8696010101 236.16550 -11.46734
    3FGL_J1544.6-1125 8696010102 236.16570 -11.46734
    3FGL_J1544.6-1125 8696010103 236.16390 -11.46833
    3FGL_J1544.6-1125 8696010104 236.16430 -11.46820
    3FGL_J1544.6-1125 8696010201 236.16440 -11.46835
Length = 39 rows

RA:236.16417, Dec -11.47222


/home/cananden/miniconda3/envs/astro/lib/python3.12/site-
packages/erfa/core.py:133: ErfaWarning: ERFA function "d2dtf" yielded 1 of
"dubious year (Note 5)"
  warn(f'ERFA function "{func_name}" yielded {wmsg}', ErfaWarning)
  ObsID    Public Date Processing Date          Epoch
---------- ----------- --------------- -----------------------
0030120101     2099_12         2018_03 2017-06-23T14:54:41.000
0030120102     2099_12         2018_03 2017-06-24T18:42:01.000
0030120103     2018_03         2018_03 2017-06-25T00:53:05.000
0030120104     2018_03         2018_03 2017-06-26T04:41:01.000
0030120105     2018_03         2018_03 2017-06-27T01:14:00.000
0030120106     2018_03         2018_03 2017-06-30T07:41:30.000
0030120107     2018_03         2018_03 2017-07-01T07:07:30.000
0030120108     2018_03         2018_03 2017-07-02T02:52:46.000
0030120109     2018_03         2018_03 2017-07-06T05:50:00.000
0030120110     2018_03         2018_03 2017-07-07T03:18:38.000
       …          …             …                     …
1030120110     2018_03         2018_03 2018-02-09T14:50:20.000
2030120101     2019_06         2019_06 2019-06-04T21:10:19.000
5030120101     2022_05         2022_05 2022-05-01T16:28:20.000
5030120102     2022_05         2022_05 2022-05-02T17:27:40.000
5030120103     2022_05         2022_05 2022-05-03T16:35:40.000
6030120101     2023_08         2023_08 2023-08-10T23:57:20.000
6030120102     2024_02         2024_02 2024-02-10T05:19:11.000
6030120103     2024_02         2024_02 2024-02-11T04:35:40.000
7030120101     2024_06         2024_06 2024-05-27T23:35:47.000
7030120102     2024_06         2024_06 2024-05-28T01:08:33.000
Length = 34 rows
Downloading
https://heasarc.gsfc.nasa.gov/FTP/nicer/data/obs/2018_01//1030120102/
Download complete.
Downloading
https://heasarc.gsfc.nasa.gov/FTP/nicer/data/obs/2018_01//1030120107/
Download complete.
```

[11]:
```
os.chdir(Path.home())
os.chdir("nicer-data")
os.chdir(f"{star_folder}")
ObsID_array = !ls -d */
ObsID_array = [d.rstrip('/') for d in ObsID_array]

print(ObsID_array)
```

```
['0030120107', '1030120102', '1030120107', '1030120108', '1030120110']
```

[4]:
```
for i in range(len(ObsID_array)):
    os.chdir(f"{ObsID_array[i]}")
    confirmation_file = f"xti/event_cl/ni{ObsID_array[i]}_0mpu7_cl.evt"
    if os.path.isfile(confirmation_file):
        print(f"{ObsID_array[i]} has been processed")
    else:
        !nicerl2 indir=. clobber=yes
    os.chdir("..")
```

```
nicerl2 1.32
-------------------------------------------------------
         Requested task operations: MKF,CALMERGE,SCREEN
       Processed prefilter_columns:
NICERV5,-MPU_OVER_COUNT,-MPU_UNDER_COUNT,-MPU_XRAY_COUNT,-MPU_ALL_COUNT
   Processed niprefilter2_coltypes: BASE,NICERV5
          Processed geomag_columns:
kp_potsdam.fits(KP),solarphi_oulu.fits(SOLAR_PHI),COR_NYM
  Inputs:
  (INDIR) .
  (UF)    ./xti/event_uf
  (CL)    ./xti/event_cl
  (MKF)   ./auxil/ni0030120107.mkf

                          MPU List: 0-6
                            OBS_ID: 0030120107
                            OBJECT: 1RXS_J154439.4-112820
                          Position: RA   236.1642  Dec   -11.4650 [deg] OBJECT
                                    RA   236.1642  Dec   -11.4660 [deg] NOMINAL
                                    RA        OBJ  Dec        OBJ [deg] REQUESTED

  CALDBVER = xti20240206
  SOFTVER  = Hea_23Feb2024_V6.33_NICER_2024-02-09_V012

========== Running nicercal ==========
nicercal 1.12
  MPU List: 0-6
  MPU0 ni0030120107_0mpu0_ufa.evt
  MPU1 ni0030120107_0mpu1_ufa.evt
```

```
  MPU2 ni0030120107_0mpu2_ufa.evt
  MPU3 ni0030120107_0mpu3_ufa.evt
  MPU4 ni0030120107_0mpu4_ufa.evt
  MPU5 ni0030120107_0mpu5_ufa.evt
  MPU6 ni0030120107_0mpu6_ufa.evt
nimpumerge 1.7
  Master sort -> ./xti/event_cl/ni0030120107_0mpu7_ufa.evt
  Append MPU GTIs
nifpmsel 1.9
  FPM List: launch
  Recalculating GTIs
========== Running niprefilter/niprefilter2 ==========
niprefilter 2.11
  Running prefilter
========== Running niprefilter2 ==========
niprefilter2 2.10
  Searching geomagnetic quantities
      (geomagnetic data from /home/cananden/nicer-
caldb/data/nicer/xti/bcf/geomag)
  Creating output file
  MPU HK files
  MPU0     (HK)     (Event)
  MPU1     (HK)     (Event)
  MPU2     (HK)     (Event)
  MPU3     (HK)     (Event)
  MPU4     (HK)     (Event)
  MPU5     (HK)     (Event)
  MPU6     (HK)     (Event)
  Event files
  Filter-level calculations
  Filter-level interpolations
    File: kp_potsdam.fits    Column: KP
    File: solarphi_oulu.fits    Column: SOLAR_PHI
  Adding COR_NYM
  Adding DELTA_SLOW_LLD
  Adding XTI_PNT_JITTER
  Adding ANG_DIST_{X,Y} (target offset)
  Appending original extension
  Filter file: ./auxil/ni0030120107.mkf
========== Running nimaketime ==========
nimaketime 1.13
        Tracking?: REQUIRED
        On Target: < 54 [arcsec]
        Elevation: < 15 [deg]
     Bright Earth: < 30 [deg]
   Number of FPMs: > 7
Star tracker good: REQUIRED
  TOT_LOWMEM_FIFO: SKIP
```

```
   DELTA_SLOW_LLD: -3.0-3.0
        NICER_SAA: SCREEN OUT
              SAA: SKIP
          COR_SAX: *-*
  UNDERONLY Range: 0-500
   OVERONLY Range: *-30
   Running maketime

                 Summary of Temporal Filtering
       Successive Filtering Stage     GoodTime #GTI
   ------------------------------   ---------- ----
            On-Target + Observing      1706.00    5
       LOWMEM + Threshold Quality      1706.00    5
                 SAA + COR + Orbit      1620.00    5
                       Undershoots      1620.00    5
                        Overshoots      1620.00    5
                      Filtered GTI      1620.00    5
========== Running nicermergeclean ==========
nicermergeclean 1.12
niautoscreen 1.3
   FPM exposure check
   FPM round-robbin check
   Under(only)shoot check
   Over(only)shoot check
   Noise rate check
   Extended noise peak (0.3-0.7 keV)
   MPU LOWMEM FIFO data losses
   Per-MPU shredded GTI check
       ---------------------------- Pointing GTIs ---------------------------
       ---------- MET ------------  ------------------UTC--------------------
   GTI        START -         STOP                START -                STOP
     1 110358549.00 - 110359245.00  2017-07-01T07:09:07 - 2017-07-01T07:20:43
     2 110362666.00 - 110362901.00  2017-07-01T08:17:44 - 2017-07-01T08:21:39
     3 110368229.00 - 110368635.00  2017-07-01T09:50:27 - 2017-07-01T09:57:13
     4 110379353.00 - 110379532.00  2017-07-01T12:55:51 - 2017-07-01T12:58:50
     5 110381326.00 - 110381510.00  2017-07-01T13:28:44 - 2017-07-01T13:31:48

                         Screened FPM / MPU quicklook table
                                DET_ID Number                        |  MPU Num.
        F0       1       2       3       4       5       6        |
   GTI  F012345670123456701234567012345670123456701234567012345670 |  M0123456
     1  F…O…O.O…O… |  M…
     2  F…O…O.O…O… |  M…
     3  F…O…O.O…O… |  M…
     4  F…O…O.O…O… |  M…
     5  F…O…O.O…O… |  M…
           Footnotes  O=known non-functional; X=screened; '.'=good
```

```
   Screening found no issues
nicerclean 1.12
           PI Range: 20 - 1500
     Trumpet Filter: 1.100000+120.000000/PI+0.000000e+00*PI**4
              Flags: Under(-) Under(-) Forced(-) NRing(-) Datamode(so+s+f)
         Expression: (((MPU_UNDER_COUNT<80)||(EVENT_FLAGS==bx0x1x000)))
  Screening…
  Updating keywords in output file…
nifpmsel 1.9
  FPM List: launch,@./xti/event_cl/ni0030120107_0mpu7_ufa.evt_detlist.txt
  Recalculating GTIs
Event files written
 UFA: ./xti/event_cl/ni0030120107_0mpu7_ufa.evt
  CL: ./xti/event_cl/ni0030120107_0mpu7_cl.evt
nicerl2 1.32
-----------------------------------------------------------
         Requested task operations: MKF,CALMERGE,SCREEN
       Processed prefilter_columns:
NICERV5,-MPU_OVER_COUNT,-MPU_UNDER_COUNT,-MPU_XRAY_COUNT,-MPU_ALL_COUNT
   Processed niprefilter2_coltypes: BASE,NICERV5
          Processed geomag_columns:
kp_potsdam.fits(KP),solarphi_oulu.fits(SOLAR_PHI),COR_NYM
  Inputs:
  (INDIR) .
  (UF)    ./xti/event_uf
  (CL)    ./xti/event_cl
  (MKF)   ./auxil/ni1030120102.mkf

                         MPU List: 0-6
                           OBS_ID: 1030120102
                           OBJECT: 1RXS_J154439.4-112820
                         Position: RA   236.1642  Dec   -11.4650 [deg] OBJECT
                                   RA   236.1636  Dec   -11.4658 [deg] NOMINAL
                                   RA        OBJ  Dec        OBJ [deg] REQUESTED

  CALDBVER = xti20240206
  SOFTVER  = Hea_23Feb2024_V6.33_NICER_2024-02-09_V012

========== Running nicercal ==========
nicercal 1.12
  MPU List: 0-6
  MPU0 ni1030120102_0mpu0_ufa.evt
  MPU1 ni1030120102_0mpu1_ufa.evt
  MPU2 ni1030120102_0mpu2_ufa.evt
  MPU3 ni1030120102_0mpu3_ufa.evt
  MPU4 ni1030120102_0mpu4_ufa.evt
  MPU5 ni1030120102_0mpu5_ufa.evt
  MPU6 ni1030120102_0mpu6_ufa.evt
```

```
nimpumerge 1.7
  Master sort -> ./xti/event_cl/ni1030120102_0mpu7_ufa.evt
  Append MPU GTIs
nifpmsel 1.9
  FPM List: launch
  Recalculating GTIs
========== Running niprefilter/niprefilter2 ==========
niprefilter 2.11
  Running prefilter
========== Running niprefilter2 ==========
niprefilter2 2.10
  Searching geomagnetic quantities
      (geomagnetic data from /home/cananden/nicer-
caldb/data/nicer/xti/bcf/geomag)
  Creating output file
  MPU HK files
  MPU0     (HK)     (Event)
  MPU1     (HK)     (Event)
  MPU2     (HK)     (Event)
  MPU3     (HK)     (Event)
  MPU4     (HK)     (Event)
  MPU5     (HK)     (Event)
  MPU6     (HK)     (Event)
  Event files
  Filter-level calculations
  Filter-level interpolations
    File: kp_potsdam.fits    Column: KP
    File: solarphi_oulu.fits    Column: SOLAR_PHI
  Adding COR_NYM
  Adding DELTA_SLOW_LLD
  Adding XTI_PNT_JITTER
  Adding ANG_DIST_{X,Y} (target offset)
  Appending original extension
  Filter file: ./auxil/ni1030120102.mkf
========== Running nimaketime ==========
nimaketime 1.13
        Tracking?: REQUIRED
        On Target: < 54 [arcsec]
        Elevation: < 15 [deg]
     Bright Earth: < 30 [deg]
   Number of FPMs: > 7
Star tracker good: REQUIRED
  TOT_LOWMEM_FIFO: SKIP
   DELTA_SLOW_LLD: -3.0-3.0
        NICER_SAA: SCREEN OUT
              SAA: SKIP
          COR_SAX: *-*
  UNDERONLY Range: 0-500
```

```
   OVERONLY Range: *-30
   Running maketime

                 Summary of Temporal Filtering
       Successive Filtering Stage     GoodTime #GTI
   ------------------------------   ---------- ----
            On-Target + Observing     28285.00   38
       LOWMEM + Threshold Quality     28285.00   38
                 SAA + COR + Orbit     24590.00   41
                       Undershoots     24590.00   41
                        Overshoots     24590.00   41
                      Filtered GTI     24534.00   29
========== Running nicermergeclean ==========
nicermergeclean 1.12
niautoscreen 1.3
   FPM exposure check
   FPM round-robbin check
   Under(only)shoot check
   Over(only)shoot check
   Noise rate check
   Extended noise peak (0.3-0.7 keV)
   MPU LOWMEM FIFO data losses
   Per-MPU shredded GTI check
       ---------------------------- Pointing GTIs ---------------------------
       ---------- MET ------------  ------------------UTC--------------------
   GTI        START -         STOP                START -                STOP
     1 128389500.00 - 128391305.00  2018-01-25T23:44:58 - 2018-01-26T00:15:03
     2 128395097.00 - 128396865.00  2018-01-26T01:18:15 - 2018-01-26T01:47:43
     3 128400663.00 - 128402425.00  2018-01-26T02:51:01 - 2018-01-26T03:20:23
     4 128406217.00 - 128407965.00  2018-01-26T04:23:35 - 2018-01-26T04:52:43
     5 128411740.00 - 128413525.00  2018-01-26T05:55:38 - 2018-01-26T06:25:23
     6 128417315.00 - 128419085.00  2018-01-26T07:28:33 - 2018-01-26T07:58:03
     7 128422900.00 - 128424645.00  2018-01-26T09:01:38 - 2018-01-26T09:30:43
     8 128428458.00 - 128430205.00  2018-01-26T10:34:16 - 2018-01-26T11:03:23
     9 128434023.00 - 128435765.00  2018-01-26T12:07:01 - 2018-01-26T12:36:03
    10 128439582.00 - 128441325.00  2018-01-26T13:39:40 - 2018-01-26T14:08:43
    11 128445138.00 - 128446885.00  2018-01-26T15:12:16 - 2018-01-26T15:41:23
    12 128450698.00 - 128452432.00  2018-01-26T16:44:56 - 2018-01-26T17:13:50
    13 128456257.00 - 128457990.00  2018-01-26T18:17:35 - 2018-01-26T18:46:28
    14 128461817.00 - 128463550.00  2018-01-26T19:50:15 - 2018-01-26T20:19:08
    15 128467377.00 - 128469111.00  2018-01-26T21:22:55 - 2018-01-26T21:51:49
    16 128472938.00 - 128474665.00  2018-01-26T22:55:36 - 2018-01-26T23:24:23

                        Screened FPM / MPU quicklook table
                               DET_ID Number                        |  MPU Num.
        F0       1       2       3       4       5       6         |
   GTI  F0123456701234567012345670123456701234567012345670123456 |  M0123456
     1  F…XO…O.O…X…O… |  M…
```

```
     2  F…XO…O.O…X…O… |  M…
     3  F…XO…O.O…X…O… |  M…
     4  F…XO…O.O…X…O… |  M…
     5  F…XO…O.O…X…O… |  M…
     6  F…XO…O.O…X…O… |  M…
     7  F…XO…O.O…X…O… |  M…
     8  F…XO…O.O…X…O… |  M…
     9  F…XO…O.O…X…O… |  M…
    10  F…XO…O.O…X…O… |  M…
    11  F…XO…O.O…X…O… |  M…
    12  F…XO…O.O…X…O… |  M…
    13  F…XO…O.O…X…O… |  M…
    14  F…XO…O.O…X…O… |  M…
    15  F…XO…O.O…X…O… |  M…
    16  F…O…O.O…X…O… |  M…
            Footnotes  O=known non-functional; X=screened; '.'=good

   Screening will disable some detectors
     detlist=10(bti:128385900-128469121),-34…
nicerclean 1.12
           PI Range: 20 - 1500
     Trumpet Filter: 1.100000+120.000000/PI+0.000000e+00*PI**4
              Flags: Under(-) Under(-) Forced(-) NRing(-) Datamode(so+s+f)
         Expression: (((MPU_UNDER_COUNT<80)||(EVENT_FLAGS==bx0x1x000)))
  Screening…
  Updating keywords in output file…
nifpmsel 1.9
  FPM List: launch,@./xti/event_cl/ni1030120102_0mpu7_ufa.evt_detlist.txt
  Recalculating GTIs
Event files written
 UFA: ./xti/event_cl/ni1030120102_0mpu7_ufa.evt
  CL: ./xti/event_cl/ni1030120102_0mpu7_cl.evt
nicerl2 1.32
-----------------------------------------------------------
         Requested task operations: SCREEN,MKF,CALMERGE
       Processed prefilter_columns:
NICERV5,-MPU_OVER_COUNT,-MPU_UNDER_COUNT,-MPU_XRAY_COUNT,-MPU_ALL_COUNT
   Processed niprefilter2_coltypes: BASE,NICERV5
          Processed geomag_columns:
kp_potsdam.fits(KP),solarphi_oulu.fits(SOLAR_PHI),COR_NYM
ERROR: uf directory ./xti/event_uf does not exist at
/home/cananden/heasoft-6.33/x86_64-pc-linux-gnu-libc2.39/bin/nicerl2 line 275.
Task nicerl2 1.32 terminating with status -1
nicerl2 1.32
-----------------------------------------------------------
         Requested task operations: CALMERGE,MKF,SCREEN
       Processed prefilter_columns:
NICERV5,-MPU_OVER_COUNT,-MPU_UNDER_COUNT,-MPU_XRAY_COUNT,-MPU_ALL_COUNT
```

```
   Processed niprefilter2_coltypes: BASE,NICERV5
          Processed geomag_columns:
kp_potsdam.fits(KP),solarphi_oulu.fits(SOLAR_PHI),COR_NYM
  Inputs:
  (INDIR) .
  (UF)    ./xti/event_uf
  (CL)    ./xti/event_cl
  (MKF)   ./auxil/ni1030120107.mkf

                        MPU List: 0-6
                          OBS_ID: 1030120107
                          OBJECT: 1RXS_J154439.4-112820
                        Position: RA   236.1642  Dec   -11.4650 [deg] OBJECT
                                  RA   236.1641  Dec   -11.4657 [deg] NOMINAL
                                  RA        OBJ  Dec        OBJ [deg] REQUESTED

  CALDBVER = xti20240206
  SOFTVER  = Hea_23Feb2024_V6.33_NICER_2024-02-09_V012

========== Running nicercal ==========
nicercal 1.12
  MPU List: 0-6
  MPU0 ni1030120107_0mpu0_ufa.evt
  MPU1 ni1030120107_0mpu1_ufa.evt
  MPU2 ni1030120107_0mpu2_ufa.evt
  MPU3 ni1030120107_0mpu3_ufa.evt
  MPU4 ni1030120107_0mpu4_ufa.evt
  MPU5 ni1030120107_0mpu5_ufa.evt
  MPU6 ni1030120107_0mpu6_ufa.evt
nimpumerge 1.7
  Master sort -> ./xti/event_cl/ni1030120107_0mpu7_ufa.evt
  Append MPU GTIs
nifpmsel 1.9
  FPM List: launch
  Recalculating GTIs
========== Running niprefilter/niprefilter2 ==========
niprefilter 2.11
  Running prefilter
========== Running niprefilter2 ==========
niprefilter2 2.10
  Searching geomagnetic quantities
      (geomagnetic data from /home/cananden/nicer-
caldb/data/nicer/xti/bcf/geomag)
  Creating output file
  MPU HK files
  MPU0     (HK)     (Event)
  MPU1     (HK)     (Event)
  MPU2     (HK)     (Event)
```

```
  MPU3     (HK)     (Event)
  MPU4     (HK)     (Event)
  MPU5     (HK)     (Event)
  MPU6     (HK)     (Event)
  Event files
  Filter-level calculations
  Filter-level interpolations
    File: kp_potsdam.fits    Column: KP
    File: solarphi_oulu.fits    Column: SOLAR_PHI
  Adding COR_NYM
  Adding DELTA_SLOW_LLD
  Adding XTI_PNT_JITTER
  Adding ANG_DIST_{X,Y} (target offset)
  Appending original extension
  Filter file: ./auxil/ni1030120107.mkf
========== Running nimaketime ==========
nimaketime 1.13
        Tracking?: REQUIRED
        On Target: < 54 [arcsec]
        Elevation: < 15 [deg]
     Bright Earth: < 30 [deg]
   Number of FPMs: > 7
Star tracker good: REQUIRED
  TOT_LOWMEM_FIFO: SKIP
   DELTA_SLOW_LLD: -3.0-3.0
        NICER_SAA: SCREEN OUT
              SAA: SKIP
          COR_SAX: *-*
  UNDERONLY Range: 0-500
   OVERONLY Range: *-30
   Running maketime

                 Summary of Temporal Filtering
       Successive Filtering Stage     GoodTime #GTI
   ------------------------------   ---------- ----
            On-Target + Observing     29039.00   24
       LOWMEM + Threshold Quality     29039.00   24
                 SAA + COR + Orbit     23668.00   27
                       Undershoots     23668.00   27
                        Overshoots     23668.00   27
                      Filtered GTI     23667.00   26
========== Running nicermergeclean ==========
nicermergeclean 1.12
niautoscreen 1.3
   FPM exposure check
   FPM round-robbin check
   Under(only)shoot check
   Over(only)shoot check
```

```
Noise rate check
Extended noise peak (0.3-0.7 keV)
MPU LOWMEM FIFO data losses
Per-MPU shredded GTI check
    ---------------------------- Pointing GTIs --------------------------
    ---------- MET ------------  ------------------UTC--------------------
GTI        START -         STOP                START -                STOP
  1 128823256.00 - 128825276.00  2018-01-31T00:14:14 - 2018-01-31T00:47:54
  2 128828819.00 - 128830836.00  2018-01-31T01:46:57 - 2018-01-31T02:20:34
  3 128834378.00 - 128836396.00  2018-01-31T03:19:36 - 2018-01-31T03:53:14
  4 128839939.00 - 128841962.00  2018-01-31T04:52:17 - 2018-01-31T05:26:00
  5 128845500.00 - 128847522.00  2018-01-31T06:24:58 - 2018-01-31T06:58:40
  6 128851059.00 - 128853082.00  2018-01-31T07:57:37 - 2018-01-31T08:31:20
  7 128856639.00 - 128858642.00  2018-01-31T09:30:37 - 2018-01-31T10:04:00
  8 128862199.00 - 128864202.00  2018-01-31T11:03:17 - 2018-01-31T11:36:40
  9 128867756.00 - 128869762.00  2018-01-31T12:35:54 - 2018-01-31T13:09:20
 10 128873319.00 - 128875512.00  2018-01-31T14:08:37 - 2018-01-31T14:45:10
 11 128878877.00 - 128881072.00  2018-01-31T15:41:15 - 2018-01-31T16:17:50
 12 128884436.00 - 128886446.00  2018-01-31T17:13:54 - 2018-01-31T17:47:24
 13 128890000.00 - 128892020.00  2018-01-31T18:46:38 - 2018-01-31T19:20:18
 14 128895559.00 - 128897792.00  2018-01-31T20:19:17 - 2018-01-31T20:56:30

                     Screened FPM / MPU quicklook table
                            DET_ID Number                        |  MPU Num.
     F0       1       2       3       4       5       6        |
GTI  F0123456701234567012345670123456701234567012345670123456 |  M0123456
  1  F…X.O…O.O…X…O… |  M…
  2  F…X.O…O.O…X…O… |  M…
  3  F…X.O…O.O…X…O… |  M…
  4  F…X.O…O.O…X…O… |  M…
  5  F…X.O…O.O…X…O… |  M…
  6  F…X.O…O.O…X…O… |  M…
  7  F…O…O.O…X…O… |  M…
  8  F…O…O.O…X…O… |  M…
  9  F…O…O.O…X…O… |  M…
 10  F…X.O…O.O…X…O… |  M…
 11  F…X.O…O.O…X…O… |  M…
 12  F…O…O.O…X…O… |  M…
 13  F…O…O.O…X…O… |  M…
 14  F…X.O…O.O…X…O… |  M…
         Footnotes  O=known non-functional; X=screened; '.'=good

Screening will disable some detectors
   detlist=07(bti:128819656-128853092|128869762-128881082|128892020-
128897802),-34…
nicerclean 1.12
          PI Range: 20 - 1500
   Trumpet Filter: 1.100000+120.000000/PI+0.000000e+00*PI**4
```

```
              Flags: Under(-) Under(-) Forced(-) NRing(-) Datamode(so+s+f)
         Expression: (((MPU_UNDER_COUNT<80)||(EVENT_FLAGS==bx0x1x000)))
  Screening…
  Updating keywords in output file…
nifpmsel 1.9
  FPM List: launch,@./xti/event_cl/ni1030120107_0mpu7_ufa.evt_detlist.txt
  Recalculating GTIs
Event files written
 UFA: ./xti/event_cl/ni1030120107_0mpu7_ufa.evt
  CL: ./xti/event_cl/ni1030120107_0mpu7_cl.evt
1030120108 has been processed
nicerl2 1.32
-----------------------------------------------------------
         Requested task operations: CALMERGE,MKF,SCREEN
       Processed prefilter_columns:
NICERV5,-MPU_OVER_COUNT,-MPU_UNDER_COUNT,-MPU_XRAY_COUNT,-MPU_ALL_COUNT
   Processed niprefilter2_coltypes: BASE,NICERV5
          Processed geomag_columns:
kp_potsdam.fits(KP),solarphi_oulu.fits(SOLAR_PHI),COR_NYM
  Inputs:
  (INDIR) .
  (UF)    ./xti/event_uf
  (CL)    ./xti/event_cl
  (MKF)   ./auxil/ni1030120110.mkf

                        MPU List: 0-6
                          OBS_ID: 1030120110
                          OBJECT: 1RXS_J154439.4-112820
                        Position: RA   236.1642  Dec   -11.4650 [deg] OBJECT
                                  RA   236.1643  Dec   -11.4659 [deg] NOMINAL
                                  RA        OBJ  Dec        OBJ [deg] REQUESTED

  CALDBVER = xti20240206
  SOFTVER  = Hea_23Feb2024_V6.33_NICER_2024-02-09_V012

========== Running nicercal ==========
nicercal 1.12
  MPU List: 0-6
  MPU0 ni1030120110_0mpu0_ufa.evt
  MPU1 ni1030120110_0mpu1_ufa.evt
  MPU2 ni1030120110_0mpu2_ufa.evt
  MPU3 ni1030120110_0mpu3_ufa.evt
  MPU4 ni1030120110_0mpu4_ufa.evt
  MPU5 ni1030120110_0mpu5_ufa.evt
  MPU6 ni1030120110_0mpu6_ufa.evt
nimpumerge 1.7
  Master sort -> ./xti/event_cl/ni1030120110_0mpu7_ufa.evt
  Append MPU GTIs
```

```
nifpmsel 1.9
  FPM List: launch
  Recalculating GTIs
========== Running niprefilter/niprefilter2 ==========
niprefilter 2.11
  Running prefilter
========== Running niprefilter2 ==========
niprefilter2 2.10
  Searching geomagnetic quantities
      (geomagnetic data from /home/cananden/nicer-
caldb/data/nicer/xti/bcf/geomag)
  Creating output file
  MPU HK files
  MPU0     (HK)     (Event)
  MPU1     (HK)     (Event)
  MPU2     (HK)     (Event)
  MPU3     (HK)     (Event)
  MPU4     (HK)     (Event)
  MPU5     (HK)     (Event)
  MPU6     (HK)     (Event)
  Event files
  Filter-level calculations
  Filter-level interpolations
    File: kp_potsdam.fits    Column: KP
    File: solarphi_oulu.fits    Column: SOLAR_PHI
  Adding COR_NYM
  Adding DELTA_SLOW_LLD
  Adding XTI_PNT_JITTER
  Adding ANG_DIST_{X,Y} (target offset)
  Appending original extension
  Filter file: ./auxil/ni1030120110.mkf
========== Running nimaketime ==========
nimaketime 1.13
        Tracking?: REQUIRED
        On Target: < 54 [arcsec]
        Elevation: < 15 [deg]
     Bright Earth: < 30 [deg]
   Number of FPMs: > 7
Star tracker good: REQUIRED
  TOT_LOWMEM_FIFO: SKIP
   DELTA_SLOW_LLD: -3.0-3.0
        NICER_SAA: SCREEN OUT
              SAA: SKIP
          COR_SAX: *-*
  UNDERONLY Range: 0-500
   OVERONLY Range: *-30
   Running maketime
```

```
                 Summary of Temporal Filtering
       Successive Filtering Stage     GoodTime #GTI
   ------------------------------   ---------- ----
            On-Target + Observing      2560.00   46
       LOWMEM + Threshold Quality      2560.00   46
                SAA + COR + Orbit      2272.00   47
                      Undershoots      2272.00   47
                       Overshoots      2272.00   47
                     Filtered GTI      2147.00   12
========== Running nicermergeclean ==========
nicermergeclean 1.12
niautoscreen 1.3
   FPM exposure check
   FPM round-robbin check
   Under(only)shoot check
   Over(only)shoot check
   Noise rate check
   Extended noise peak (0.3-0.7 keV)
   MPU LOWMEM FIFO data losses
   Per-MPU shredded GTI check
       ---------------------------- Pointing GTIs ---------------------------
       ---------- MET ------------  -------------------UTC--------------------
   GTI        START -         STOP                START -                STOP
     1 129653489.00 - 129654272.00  2018-02-09T14:51:27 - 2018-02-09T15:04:30
     2 129659067.00 - 129659831.00  2018-02-09T16:24:25 - 2018-02-09T16:37:09
     3 129664628.00 - 129665393.00  2018-02-09T17:57:06 - 2018-02-09T18:09:51
     4 129676243.00 - 129676525.00  2018-02-09T21:10:41 - 2018-02-09T21:15:23

                        Screened FPM / MPU quicklook table
                               DET_ID Number                        |  MPU Num.
        F0       1       2       3       4       5       6        |
   GTI  F0123456701234567012345670123456701234567012345670123456 |  M0123456
     1  F…O…O.O…X..X…X…O… |  M…
     2  F…O…O.O…X..X…X…O… |  M…
     3  F…O…O.O…X..X…X…O… |  M…
     4  F…O…O.O…X…X…O… |  M…
            Footnotes  O=known non-functional; X=screened; '.'=good

   Screening will disable some detectors
     detlist=-34,37(bti:129649889-129665403),-47…
nicerclean 1.12
           PI Range: 20 - 1500
     Trumpet Filter: 1.100000+120.000000/PI+0.000000e+00*PI**4
              Flags: Under(-) Under(-) Forced(-) NRing(-) Datamode(so+s+f)
         Expression: (((MPU_UNDER_COUNT<80)||(EVENT_FLAGS==bx0x1x000)))
  Screening…
  Updating keywords in output file…
nifpmsel 1.9
```

```
  FPM List: launch,@./xti/event_cl/ni1030120110_0mpu7_ufa.evt_detlist.txt
  Recalculating GTIs
Event files written
 UFA: ./xti/event_cl/ni1030120110_0mpu7_ufa.evt
  CL: ./xti/event_cl/ni1030120110_0mpu7_cl.evt
```

[5]:
```python
os.chdir("/home/cananden/nicer-data/")
os.chdir(f"{star_folder}")

for i in range(len(ObsID_array)):
    os.chdir(f"{ObsID_array[i]}")
    os.chdir("xti")

    if os.path.isfile(f"{ObsID_array[i]}.fits"):
        print(f"{ObsID_array[i]}.fits already generated")
        os.chdir("..")
        os.chdir("..")
    else:
        cmd = f"""
xselect << EOF
nicer
no
read events
./event_cl/
ni{ObsID_array[i]}_0mpu7_cl.evt
yes
filter pha_cut {filter_min} {filter_max}
extract curve
extract curve exposure=0.0
save curve {ObsID_array[i]}.fits
exit
EOF
"""
        subprocess.run(cmd, shell=True, executable="/bin/bash")
        os.chdir("..")
        os.chdir("..")
```

```
                    **  XSELECT V2.5b  **

> Enter session name >[nicer] nicer
nicer:SUZAKU > no
 Command not found; type ? for a command listing
nicer:SUZAKU > read events
> Enter the Event file dir >[./event_cl/] ./event_cl/
> Enter Event file list >[ni1030120108_0mpu7_cl_bary.evt]
ni0030120107_0mpu7_cl.evt
Got new mission: NICER
```

```
> Reset the mission ? >[yes] yes


        OBJECT                OBS_ID      DATE-OBS
      1 1RXS_J154439.4-112820 0030120107  2017-07-01T07:07:30

 Notes: XSELECT set up for      NICER
 Time keyword is TIME       in units of s
 Default timing binsize =   16.000

Setting…
 Image  keywords   = RAWX       RAWY        with binning =    1
 Energy keyword   = PI                      with binning =    1

Getting Min and Max for Energy Column…
Got min and max for PI:     0   1500

Got the minimum time resolution of the read data:  0.40000E-07
MJDREF =  5.6658000777593E+04 with TIMESYS = TT
 Number of files read in:            1

******************** Observation Catalogue ********************

Data Directory is: /home/cananden/nicer-
data/1RXS_J154439.4-112820/0030120107/xti/event_cl/
HK Directory is: /home/cananden/nicer-
data/1RXS_J154439.4-112820/0030120107/xti/event_cl/

nicer:NICER-XTI-PHOTON > filter pha_cut 30 1200
nicer:NICER-XTI-PHOTON > extract curve
extractor v6.11    12 Dec 2023
 Getting FITS WCS Keywords
 Doing file: /home/cananden/nicer-
data/1RXS_J154439.4-112820/0030120107/xti/event_cl/ni0030120107_0mpu7_cl.evt
100% completed
          Total      Good    Bad: Time     Phase     Grade       Cut
           4839      4427            0         0         0       412
===============================================================================
    Grand Total      Good    Bad: Time     Phase     Grade       Cut
           4839      4427            0         0         0       412
   in  1619.2     seconds
 Fits light curve has   4427.     counts for   2.734     counts/sec
 Thresholding removed more than half the bins
 Try exposure=0.0 on the extract command in xselect
 or lcthresh=0.0 if running extractor stand-alone
nicer:NICER-XTI-PHOTON > extract curve exposure=0.0
extractor v6.11    12 Dec 2023
 Getting FITS WCS Keywords
```

```
 Doing file: /home/cananden/nicer-
data/1RXS_J154439.4-112820/0030120107/xti/event_cl/ni0030120107_0mpu7_cl.evt
100% completed
          Total      Good    Bad: Time     Phase     Grade       Cut
           4839      4427            0         0         0       412
================================================================================
    Grand Total      Good    Bad: Time     Phase     Grade       Cut
           4839      4427            0         0         0       412
   in  1619.2     seconds
 Fits light curve has   4427.     counts for   2.734     counts/sec
 Thresholding removed more than half the bins
 Try exposure=0.0 on the extract command in xselect
 or lcthresh=0.0 if running extractor stand-alone
nicer:NICER-XTI-PHOTON > save curve 0030120107.fits
Wrote FITS light curve to file 0030120107.fits
nicer:NICER-XTI-PHOTON > exit
> Save this session? >[n]
                        **  XSELECT V2.5b  **

> Enter session name >[nicer] nicer
nicer:SUZAKU > no
 Command not found; type ? for a command listing
nicer:SUZAKU > read events
> Enter the Event file dir >[./event_cl/] ./event_cl/
> Enter Event file list >[ni0030120107_0mpu7_cl.evt] ni1030120102_0mpu7_cl.evt
Got new mission: NICER
> Reset the mission ? >[yes] yes


        OBJECT                OBS_ID      DATE-OBS
      1 1RXS_J154439.4-112820 1030120102  2018-01-25T23:43:00

 Notes: XSELECT set up for      NICER
 Time keyword is TIME       in units of s
 Default timing binsize =   16.000

Setting…
 Image  keywords   = RAWX       RAWY        with binning =    1
 Energy keyword   = PI                      with binning =    1

Getting Min and Max for Energy Column…
Got min and max for PI:     0   1500

Got the minimum time resolution of the read data:  0.40000E-07
MJDREF =  5.6658000777593E+04 with TIMESYS = TT
 Number of files read in:            1

******************** Observation Catalogue ********************
```

```
Data Directory is: /home/cananden/nicer-
data/1RXS_J154439.4-112820/1030120102/xti/event_cl/
HK Directory is: /home/cananden/nicer-
data/1RXS_J154439.4-112820/1030120102/xti/event_cl/

nicer:NICER-XTI-PHOTON > filter pha_cut 30 1200
nicer:NICER-XTI-PHOTON > extract curve
extractor v6.11    12 Dec 2023
 Getting FITS WCS Keywords
 Doing file: /home/cananden/nicer-
data/1RXS_J154439.4-112820/1030120102/xti/event_cl/ni1030120102_0mpu7_cl.evt
100% completed
          Total      Good    Bad: Time     Phase     Grade       Cut
          61513     48009            0         0         0     13504
================================================================================
    Grand Total      Good    Bad: Time     Phase     Grade       Cut
          61513     48009            0         0         0     13504
   in  24534.     seconds
 Fits light curve has  4.8009E+04 counts for   1.957     counts/sec
 Thresholding removed more than half the bins
 Try exposure=0.0 on the extract command in xselect
 or lcthresh=0.0 if running extractor stand-alone
nicer:NICER-XTI-PHOTON > extract curve exposure=0.0
extractor v6.11    12 Dec 2023
 Getting FITS WCS Keywords
 Doing file: /home/cananden/nicer-
data/1RXS_J154439.4-112820/1030120102/xti/event_cl/ni1030120102_0mpu7_cl.evt
100% completed
          Total      Good    Bad: Time     Phase     Grade       Cut
          61513     48009            0         0         0     13504
================================================================================
    Grand Total      Good    Bad: Time     Phase     Grade       Cut
          61513     48009            0         0         0     13504
   in  24534.     seconds
 Fits light curve has  4.8009E+04 counts for   1.957     counts/sec
 Thresholding removed more than half the bins
 Try exposure=0.0 on the extract command in xselect
 or lcthresh=0.0 if running extractor stand-alone
nicer:NICER-XTI-PHOTON > save curve 1030120102.fits
Wrote FITS light curve to file 1030120102.fits
nicer:NICER-XTI-PHOTON > exit
> Save this session? >[n]
                        **  XSELECT V2.5b  **

> Enter session name >[nicer] nicer
nicer:SUZAKU > no
 Command not found; type ? for a command listing
```

```
nicer:SUZAKU > read events
> Enter the Event file dir >[./event_cl/] ./event_cl/
The directory ./event_cl/ doesn't exist
nicer:SUZAKU > ni1030120106_0mpu7_cl.evt
 Command not found; type ? for a command listing
nicer:SUZAKU > yes
 Command not found; type ? for a command listing
nicer:SUZAKU > filter pha_cut 30 1200
nicer:SUZAKU > extract curve
No data has been read in yet
 XSL_BIN: Status =    -5
nicer:SUZAKU > extract curve exposure=0.0
No data has been read in yet
 XSL_BIN: Status =    -5
nicer:SUZAKU > save curve 1030120106.fits
 No curve has been accumulated yet.
nicer:SUZAKU > exit
> Save this session? >[n]
                        **  XSELECT V2.5b  **

> Enter session name >[nicer] nicer
nicer:SUZAKU > no
 Command not found; type ? for a command listing
nicer:SUZAKU > read events
> Enter the Event file dir >[./event_cl/] ./event_cl/
> Enter Event file list >[ni1030120102_0mpu7_cl.evt] ni1030120107_0mpu7_cl.evt
Got new mission: NICER
> Reset the mission ? >[yes] yes


        OBJECT                OBS_ID      DATE-OBS
      1 1RXS_J154439.4-112820 1030120107  2018-01-31T00:11:24

 Notes: XSELECT set up for      NICER
 Time keyword is TIME       in units of s
 Default timing binsize =   16.000

Setting…
 Image  keywords   = RAWX       RAWY        with binning =    1
 Energy keyword   = PI                      with binning =    1

Getting Min and Max for Energy Column…
Got min and max for PI:     0   1500

Got the minimum time resolution of the read data:  0.40000E-07
MJDREF =  5.6658000777593E+04 with TIMESYS = TT
 Number of files read in:            1
```

```
******************** Observation Catalogue ********************

Data Directory is: /home/cananden/nicer-
data/1RXS_J154439.4-112820/1030120107/xti/event_cl/
HK Directory is: /home/cananden/nicer-
data/1RXS_J154439.4-112820/1030120107/xti/event_cl/

nicer:NICER-XTI-PHOTON > filter pha_cut 30 1200
nicer:NICER-XTI-PHOTON > extract curve
extractor v6.11    12 Dec 2023
 Getting FITS WCS Keywords
 Doing file: /home/cananden/nicer-
data/1RXS_J154439.4-112820/1030120107/xti/event_cl/ni1030120107_0mpu7_cl.evt
100% completed
          Total      Good    Bad: Time     Phase     Grade       Cut
          59321     47714            0         0         0     11607
================================================================================
    Grand Total      Good    Bad: Time     Phase     Grade       Cut
          59321     47714            0         0         0     11607
   in  23667.     seconds
 Fits light curve has  4.7714E+04 counts for   2.016     counts/sec
 Thresholding removed more than half the bins
 Try exposure=0.0 on the extract command in xselect
 or lcthresh=0.0 if running extractor stand-alone
nicer:NICER-XTI-PHOTON > extract curve exposure=0.0
extractor v6.11    12 Dec 2023
 Getting FITS WCS Keywords
 Doing file: /home/cananden/nicer-
data/1RXS_J154439.4-112820/1030120107/xti/event_cl/ni1030120107_0mpu7_cl.evt
100% completed
          Total      Good    Bad: Time     Phase     Grade       Cut
          59321     47714            0         0         0     11607
================================================================================
    Grand Total      Good    Bad: Time     Phase     Grade       Cut
          59321     47714            0         0         0     11607
   in  23667.     seconds
 Fits light curve has  4.7714E+04 counts for   2.016     counts/sec
 Thresholding removed more than half the bins
 Try exposure=0.0 on the extract command in xselect
 or lcthresh=0.0 if running extractor stand-alone
nicer:NICER-XTI-PHOTON > save curve 1030120107.fits
Wrote FITS light curve to file 1030120107.fits
nicer:NICER-XTI-PHOTON > exit
> Save this session? >[n] 1030120108.fits already generated

                        **  XSELECT V2.5b  **

> Enter session name >[nicer] nicer
```

```
nicer:SUZAKU > no
 Command not found; type ? for a command listing
nicer:SUZAKU > read events
> Enter the Event file dir >[./event_cl/] ./event_cl/
> Enter Event file list >[ni1030120107_0mpu7_cl.evt] ni1030120110_0mpu7_cl.evt
Got new mission: NICER
> Reset the mission ? >[yes] yes


        OBJECT                OBS_ID      DATE-OBS
      1 1RXS_J154439.4-112820 1030120110  2018-02-09T14:50:20

 Notes: XSELECT set up for      NICER
 Time keyword is TIME       in units of s
 Default timing binsize =   16.000

Setting…
 Image  keywords   = RAWX       RAWY        with binning =    1
 Energy keyword   = PI                      with binning =    1

Getting Min and Max for Energy Column…
Got min and max for PI:     0   1500

Got the minimum time resolution of the read data:  0.40000E-07
MJDREF =  5.6658000777593E+04 with TIMESYS = TT
 Number of files read in:            1

******************** Observation Catalogue ********************

Data Directory is: /home/cananden/nicer-
data/1RXS_J154439.4-112820/1030120110/xti/event_cl/
HK Directory is: /home/cananden/nicer-
data/1RXS_J154439.4-112820/1030120110/xti/event_cl/

nicer:NICER-XTI-PHOTON > filter pha_cut 30 1200
nicer:NICER-XTI-PHOTON > extract curve
extractor v6.11    12 Dec 2023
 Getting FITS WCS Keywords
 Doing file: /home/cananden/nicer-
data/1RXS_J154439.4-112820/1030120110/xti/event_cl/ni1030120110_0mpu7_cl.evt
100% completed
          Total      Good    Bad: Time     Phase     Grade       Cut
           4702      3890            0         0         0       812
================================================================================
    Grand Total      Good    Bad: Time     Phase     Grade       Cut
           4702      3890            0         0         0       812
   in  2147.0     seconds
 Fits light curve has   3890.     counts for   1.812     counts/sec
```

```
 Thresholding removed more than half the bins
 Try exposure=0.0 on the extract command in xselect
 or lcthresh=0.0 if running extractor stand-alone
nicer:NICER-XTI-PHOTON > extract curve exposure=0.0
extractor v6.11    12 Dec 2023
 Getting FITS WCS Keywords
 Doing file: /home/cananden/nicer-
data/1RXS_J154439.4-112820/1030120110/xti/event_cl/ni1030120110_0mpu7_cl.evt
100% completed
          Total      Good    Bad: Time     Phase     Grade       Cut
           4702      3890            0         0         0       812
================================================================================
    Grand Total      Good    Bad: Time     Phase     Grade       Cut
           4702      3890            0         0         0       812
   in  2147.0     seconds
 Fits light curve has   3890.     counts for   1.812     counts/sec
 Thresholding removed more than half the bins
 Try exposure=0.0 on the extract command in xselect
 or lcthresh=0.0 if running extractor stand-alone
nicer:NICER-XTI-PHOTON > save curve 1030120110.fits
Wrote FITS light curve to file 1030120110.fits
nicer:NICER-XTI-PHOTON > exit
> Save this session? >[n]
```

[6]:
```python
for i in range(len(ObsID_array)):
    os.chdir(f"/home/cananden/nicer-data/{star_folder}/{ObsID_array[i]}")

    infile = f"xti/event_cl/ni{ObsID_array[i]}_0mpu7_cl.evt"
    outfile = f"xti/event_cl/ni{ObsID_array[i]}_0mpu7_cl_bary.evt"
    orbitf = f"auxil/ni{ObsID_array[i]}.orb.gz"

    if os.path.isfile(outfile):
        os.chdir("..")
        print(f"ni{ObsID_array[i]}_0mpu7_cl_bary.evt already exists")
        continue

    cmd_bary = [
        "barycorr",
        f"infile={infile}",
        f"outfile={outfile}",
        f"orbitfiles={orbitf}",
        f"ra={RA}",
        f"dec={Dec}",
        "refframe=ICRS",
        "ephem=JPLEPH.430",
        "barytime=no",
        "clobber=yes"
```

```
    ]

    env = os.environ.copy()
    env["HEADASNOQUERY"] = "1"
    env["HEADASYES"] = "1"
    subprocess.run(cmd_bary, env=env)
    os.chdir("..")
```

```
**** Running barycorr v2.17 ****
barycorr: shell ftcopy infile='xti/event_cl/ni0030120107_0mpu7_cl.evt'
outfile='xti/event_cl/ni0030120107_0mpu7_cl_bary.evtworkfile19975' copyall=YES
clobber=YES chatter=0 history=NO
barycorr: running hdaxbary -i 'auxil/ni0030120107.orb.gz' -f
xti/event_cl/ni0030120107_0mpu7_cl_bary.evtworkfile19975 -jpleph 430 -ra
236.16417 -dec -11.47222 -ref ICRS
barycorr: shell hdaxbary -i 'auxil/ni0030120107.orb.gz' -f
xti/event_cl/ni0030120107_0mpu7_cl_bary.evtworkfile19975 -jpleph 430 -ra
236.16417 -dec -11.47222 -ref ICRS 2>&1
===> No barycenter correction applied to HDU 0
     TIMESYS is
<=== Finished after 11 HDUs
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430barycorr: exited with status 0
barycorr: shell ftchecksum
infile='xti/event_cl/ni0030120107_0mpu7_cl_bary.evtworkfile19975' update=yes
datasum=YES chatter=0
barycorr: shell ftcopy
infile='xti/event_cl/ni0030120107_0mpu7_cl_bary.evtworkfile19975'
outfile='xti/event_cl/ni0030120107_0mpu7_cl_bary.evt' copyall=YES clobber=YES
chatter=0 history=NO
barycorr: barycorr successfully completed

**** Running barycorr v2.17 ****
barycorr: shell ftcopy infile='xti/event_cl/ni1030120102_0mpu7_cl.evt'
outfile='xti/event_cl/ni1030120102_0mpu7_cl_bary.evtworkfile19992' copyall=YES
clobber=YES chatter=0 history=NO
barycorr: running hdaxbary -i 'auxil/ni1030120102.orb.gz' -f
xti/event_cl/ni1030120102_0mpu7_cl_bary.evtworkfile19992 -jpleph 430 -ra
236.16417 -dec -11.47222 -ref ICRS
```

```
barycorr: shell hdaxbary -i 'auxil/ni1030120102.orb.gz' -f
xti/event_cl/ni1030120102_0mpu7_cl_bary.evtworkfile19992 -jpleph 430 -ra
236.16417 -dec -11.47222 -ref ICRS 2>&1
===> No barycenter correction applied to HDU 0
     TIMESYS is
<=== Finished after 12 HDUs
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430barycorr: exited with status 0
barycorr: shell ftchecksum
infile='xti/event_cl/ni1030120102_0mpu7_cl_bary.evtworkfile19992' update=yes
datasum=YES chatter=0
barycorr: shell ftcopy
infile='xti/event_cl/ni1030120102_0mpu7_cl_bary.evtworkfile19992'
outfile='xti/event_cl/ni1030120102_0mpu7_cl_bary.evt' copyall=YES clobber=YES
chatter=0 history=NO
barycorr: barycorr successfully completed

**** Running barycorr v2.17 ****
barycorr: shell ftcopy infile='xti/event_cl/ni1030120107_0mpu7_cl.evt'
outfile='xti/event_cl/ni1030120107_0mpu7_cl_bary.evtworkfile20012' copyall=YES
clobber=YES chatter=0 history=NO
barycorr: running hdaxbary -i 'auxil/ni1030120107.orb.gz' -f
xti/event_cl/ni1030120107_0mpu7_cl_bary.evtworkfile20012 -jpleph 430 -ra
236.16417 -dec -11.47222 -ref ICRS
barycorr: shell hdaxbary -i 'auxil/ni1030120107.orb.gz' -f
xti/event_cl/ni1030120107_0mpu7_cl_bary.evtworkfile20012 -jpleph 430 -ra
236.16417 -dec -11.47222 -ref ICRS 2>&1
===> No barycenter correction applied to HDU 0
     TIMESYS is
<=== Finished after 12 HDUs
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430
```

```
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430barycorr: exited with status 0
barycorr: shell ftchecksum
infile='xti/event_cl/ni1030120107_0mpu7_cl_bary.evtworkfile20012' update=yes
datasum=YES chatter=0
barycorr: shell ftcopy
infile='xti/event_cl/ni1030120107_0mpu7_cl_bary.evtworkfile20012'
outfile='xti/event_cl/ni1030120107_0mpu7_cl_bary.evt' copyall=YES clobber=YES
chatter=0 history=NO
barycorr: barycorr successfully completed

ni1030120108_0mpu7_cl_bary.evt already exitst

Parameter infile: file "xti/event_cl/ni1030120106_0mpu7_cl.evt" is not
accessible in mode r.
error getting infile parameter at /home/cananden/heasoft-6.33/x86_64-pc-linux-
gnu-libc2.39/bin/barycorr line 144.
ERROR: Operation not permitted
Task barycorr 0.0 terminating with status 1

**** Running barycorr v2.17 ****
barycorr: shell ftcopy infile='xti/event_cl/ni1030120110_0mpu7_cl.evt'
outfile='xti/event_cl/ni1030120110_0mpu7_cl_bary.evtworkfile20027' copyall=YES
clobber=YES chatter=0 history=NO
barycorr: running hdaxbary -i 'auxil/ni1030120110.orb.gz' -f
xti/event_cl/ni1030120110_0mpu7_cl_bary.evtworkfile20027 -jpleph 430 -ra
236.16417 -dec -11.47222 -ref ICRS
barycorr: shell hdaxbary -i 'auxil/ni1030120110.orb.gz' -f
xti/event_cl/ni1030120110_0mpu7_cl_bary.evtworkfile20027 -jpleph 430 -ra
236.16417 -dec -11.47222 -ref ICRS 2>&1
===> No barycenter correction applied to HDU 0
     TIMESYS is
<=== Finished after 12 HDUs
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430
hdaxbary: Using JPL Planetary Ephemeris DE-430barycorr: exited with status 0
barycorr: shell ftchecksum
infile='xti/event_cl/ni1030120110_0mpu7_cl_bary.evtworkfile20027' update=yes
datasum=YES chatter=0
barycorr: shell ftcopy
infile='xti/event_cl/ni1030120110_0mpu7_cl_bary.evtworkfile20027'
```

```
outfile='xti/event_cl/ni1030120110_0mpu7_cl_bary.evt' copyall=YES clobber=YES
chatter=0 history=NO
barycorr: barycorr successfully completed
```

[7]:
```
os.chdir("/home/cananden/nicer-data/")
os.chdir(f"{star_folder}")

for i in range(len(ObsID_array)):
    os.chdir(f"{ObsID_array[i]}")
    os.chdir("xti")

    if os.path.isfile(f"{ObsID_array[i]}_bary.fits"):
        print(f"{ObsID_array[i]}_bary.fits already generated")
        os.chdir("..")
        os.chdir("..")
        continue
    else:
        cmd = f"""
xselect << EOF
nicer
no
read events
./event_cl/
ni{ObsID_array[i]}_0mpu7_cl_bary.evt
yes
extract curve
extract curve exposure=0.0
save curve {ObsID_array[i]}_bary.fits
exit
EOF
"""
        subprocess.run(cmd, shell=True, executable="/bin/bash")
        os.chdir("..")
        os.chdir("..")
```

```
                    **  XSELECT V2.5b  **

> Enter session name >[nicer] nicer
nicer:SUZAKU > no
 Command not found; type ? for a command listing
nicer:SUZAKU > read events
> Enter the Event file dir >[./event_cl/] ./event_cl/
> Enter Event file list >[ni1030120110_0mpu7_cl.evt]
ni0030120107_0mpu7_cl_bary.evt
Got new mission: NICER
> Reset the mission ? >[yes] yes
```

```
        OBJECT                OBS_ID      DATE-OBS
      1 1RXS_J154439.4-112820 0030120107  2017-07-01T07:07:30

 Notes: XSELECT set up for      NICER
 Time keyword is TIME       in units of s
 Default timing binsize =   16.000

Setting…
 Image  keywords   = RAWX       RAWY        with binning =    1
 Energy keyword   = PI                      with binning =    1

Getting Min and Max for Energy Column…
Got min and max for PI:     0   1500

Got the minimum time resolution of the read data:  0.40000E-07
MJDREF =  5.6658000777593E+04 with TIMESYS = TDB
 Number of files read in:            1

******************** Observation Catalogue ********************

Data Directory is: /home/cananden/nicer-
data/1RXS_J154439.4-112820/0030120107/xti/event_cl/
HK Directory is: /home/cananden/nicer-
data/1RXS_J154439.4-112820/0030120107/xti/event_cl/

nicer:NICER-XTI-PHOTON > extract curve
extractor v6.11    12 Dec 2023
 Getting FITS WCS Keywords
 Doing file: /home/cananden/nicer-data/1RXS_J154439.4-
112820/0030120107/xti/event_cl/ni0030120107_0mpu7_cl_bary.evt
100% completed
          Total      Good    Bad: Time     Phase     Grade       Cut
           4839      4839            0         0         0         0
================================================================================
    Grand Total      Good    Bad: Time     Phase     Grade       Cut
           4839      4839            0         0         0         0
   in  1619.1     seconds
 Fits light curve has   4839.     counts for   2.989     counts/sec
 Thresholding removed more than half the bins
 Try exposure=0.0 on the extract command in xselect
 or lcthresh=0.0 if running extractor stand-alone
nicer:NICER-XTI-PHOTON > extract curve exposure=0.0
extractor v6.11    12 Dec 2023
 Getting FITS WCS Keywords
 Doing file: /home/cananden/nicer-data/1RXS_J154439.4-
112820/0030120107/xti/event_cl/ni0030120107_0mpu7_cl_bary.evt
```

```
100% completed
          Total      Good    Bad: Time     Phase     Grade       Cut
           4839      4839            0         0         0         0
================================================================================
    Grand Total      Good    Bad: Time     Phase     Grade       Cut
           4839      4839            0         0         0         0
   in  1619.1     seconds
 Fits light curve has   4839.     counts for   2.989     counts/sec
 Thresholding removed more than half the bins
 Try exposure=0.0 on the extract command in xselect
 or lcthresh=0.0 if running extractor stand-alone
nicer:NICER-XTI-PHOTON > save curve 0030120107_bary.fits
Wrote FITS light curve to file 0030120107_bary.fits
nicer:NICER-XTI-PHOTON > exit
> Save this session? >[n]
                        **  XSELECT V2.5b  **

> Enter session name >[nicer] nicer
nicer:SUZAKU > no
 Command not found; type ? for a command listing
nicer:SUZAKU > read events
> Enter the Event file dir >[./event_cl/] ./event_cl/
> Enter Event file list >[ni0030120107_0mpu7_cl_bary.evt]
ni1030120102_0mpu7_cl_bary.evt
Got new mission: NICER
> Reset the mission ? >[yes] yes


        OBJECT                OBS_ID      DATE-OBS
      1 1RXS_J154439.4-112820 1030120102  2018-01-25T23:43:00

 Notes: XSELECT set up for      NICER
 Time keyword is TIME       in units of s
 Default timing binsize =   16.000

Setting…
 Image  keywords   = RAWX       RAWY        with binning =    1
 Energy keyword   = PI                      with binning =    1

Getting Min and Max for Energy Column…
Got min and max for PI:     0   1500

Got the minimum time resolution of the read data:  0.40000E-07
MJDREF =  5.6658000777593E+04 with TIMESYS = TDB
 Number of files read in:            1

******************** Observation Catalogue ********************
```

```
Data Directory is: /home/cananden/nicer-
data/1RXS_J154439.4-112820/1030120102/xti/event_cl/
HK Directory is: /home/cananden/nicer-
data/1RXS_J154439.4-112820/1030120102/xti/event_cl/

nicer:NICER-XTI-PHOTON > extract curve
extractor v6.11    12 Dec 2023
 Getting FITS WCS Keywords
 Doing file: /home/cananden/nicer-data/1RXS_J154439.4-
112820/1030120102/xti/event_cl/ni1030120102_0mpu7_cl_bary.evt
100% completed
          Total      Good    Bad: Time     Phase     Grade       Cut
          61513     61513            0         0         0         0
================================================================================
    Grand Total      Good    Bad: Time     Phase     Grade       Cut
          61513     61513            0         0         0         0
   in  24537.     seconds
 Fits light curve has  6.1513E+04 counts for   2.507     counts/sec
 Thresholding removed more than half the bins
 Try exposure=0.0 on the extract command in xselect
 or lcthresh=0.0 if running extractor stand-alone
nicer:NICER-XTI-PHOTON > extract curve exposure=0.0
extractor v6.11    12 Dec 2023
 Getting FITS WCS Keywords
 Doing file: /home/cananden/nicer-data/1RXS_J154439.4-
112820/1030120102/xti/event_cl/ni1030120102_0mpu7_cl_bary.evt
100% completed
          Total      Good    Bad: Time     Phase     Grade       Cut
          61513     61513            0         0         0         0
================================================================================
    Grand Total      Good    Bad: Time     Phase     Grade       Cut
          61513     61513            0         0         0         0
   in  24537.     seconds
 Fits light curve has  6.1513E+04 counts for   2.507     counts/sec
 Thresholding removed more than half the bins
 Try exposure=0.0 on the extract command in xselect
 or lcthresh=0.0 if running extractor stand-alone
nicer:NICER-XTI-PHOTON > save curve 1030120102_bary.fits
Wrote FITS light curve to file 1030120102_bary.fits
nicer:NICER-XTI-PHOTON > exit
> Save this session? >[n]
                         **  XSELECT V2.5b  **

> Enter session name >[nicer] nicer
nicer:SUZAKU > no
 Command not found; type ? for a command listing
nicer:SUZAKU > read events
> Enter the Event file dir >[./event_cl/] ./event_cl/
```

```
The directory ./event_cl/ doesn't exist
nicer:SUZAKU > ni1030120106_0mpu7_cl_bary.evt
 Command not found; type ? for a command listing
nicer:SUZAKU > yes
 Command not found; type ? for a command listing
nicer:SUZAKU > extract curve
No data has been read in yet
 XSL_BIN: Status =    -5
nicer:SUZAKU > extract curve exposure=0.0
No data has been read in yet
 XSL_BIN: Status =    -5
nicer:SUZAKU > save curve 1030120106_bary.fits
 No curve has been accumulated yet.
nicer:SUZAKU > exit
> Save this session? >[n]
                         **  XSELECT V2.5b  **

> Enter session name >[nicer] nicer
nicer:SUZAKU > no
 Command not found; type ? for a command listing
nicer:SUZAKU > read events
> Enter the Event file dir >[./event_cl/] ./event_cl/
> Enter Event file list >[ni1030120102_0mpu7_cl_bary.evt]
ni1030120107_0mpu7_cl_bary.evt
Got new mission: NICER
> Reset the mission ? >[yes] yes


        OBJECT                OBS_ID      DATE-OBS
      1 1RXS_J154439.4-112820 1030120107  2018-01-31T00:11:24

 Notes: XSELECT set up for      NICER
 Time keyword is TIME       in units of s
 Default timing binsize =   16.000

Setting…
 Image  keywords   = RAWX       RAWY        with binning =    1
 Energy keyword   = PI                      with binning =    1

Getting Min and Max for Energy Column…
Got min and max for PI:     0   1500

Got the minimum time resolution of the read data:  0.40000E-07
MJDREF =  5.6658000777593E+04 with TIMESYS = TDB
 Number of files read in:            1

******************** Observation Catalogue ********************
```

```
Data Directory is: /home/cananden/nicer-
data/1RXS_J154439.4-112820/1030120107/xti/event_cl/
HK Directory is: /home/cananden/nicer-
data/1RXS_J154439.4-112820/1030120107/xti/event_cl/

nicer:NICER-XTI-PHOTON > extract curve
extractor v6.11    12 Dec 2023
 Getting FITS WCS Keywords
 Doing file: /home/cananden/nicer-data/1RXS_J154439.4-
112820/1030120107/xti/event_cl/ni1030120107_0mpu7_cl_bary.evt
100% completed
          Total      Good    Bad: Time     Phase     Grade       Cut
          59321     59321            0         0         0         0
================================================================================
    Grand Total      Good    Bad: Time     Phase     Grade       Cut
          59321     59321            0         0         0         0
   in  23669.     seconds
 Fits light curve has  5.9321E+04 counts for   2.506     counts/sec
 Thresholding removed more than half the bins
 Try exposure=0.0 on the extract command in xselect
 or lcthresh=0.0 if running extractor stand-alone
nicer:NICER-XTI-PHOTON > extract curve exposure=0.0
extractor v6.11    12 Dec 2023
 Getting FITS WCS Keywords
 Doing file: /home/cananden/nicer-data/1RXS_J154439.4-
112820/1030120107/xti/event_cl/ni1030120107_0mpu7_cl_bary.evt
100% completed
          Total      Good    Bad: Time     Phase     Grade       Cut
          59321     59321            0         0         0         0
================================================================================
    Grand Total      Good    Bad: Time     Phase     Grade       Cut
          59321     59321            0         0         0         0
   in  23669.     seconds
 Fits light curve has  5.9321E+04 counts for   2.506     counts/sec
 Thresholding removed more than half the bins
 Try exposure=0.0 on the extract command in xselect
 or lcthresh=0.0 if running extractor stand-alone
nicer:NICER-XTI-PHOTON > save curve 1030120107_bary.fits
Wrote FITS light curve to file 1030120107_bary.fits
nicer:NICER-XTI-PHOTON > exit
> Save this session? >[n] 1030120108_bary.fits already generated

                        **  XSELECT V2.5b  **

> Enter session name >[nicer] nicer
nicer:SUZAKU > no
 Command not found; type ? for a command listing
nicer:SUZAKU > read events
```

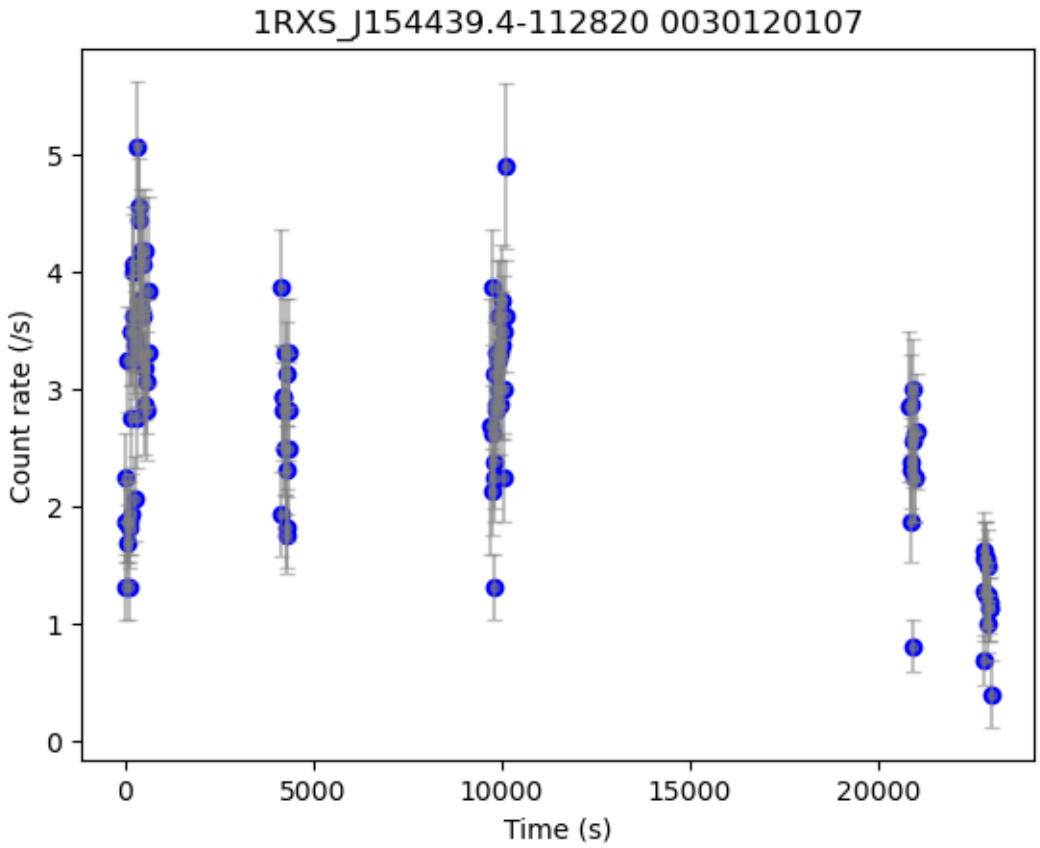
```
> Enter the Event file dir >[./event_cl/] ./event_cl/
> Enter Event file list >[ni1030120107_0mpu7_cl_bary.evt]
ni1030120110_0mpu7_cl_bary.evt
Got new mission: NICER
> Reset the mission ? >[yes] yes


        OBJECT                OBS_ID      DATE-OBS
      1 1RXS_J154439.4-112820 1030120110  2018-02-09T14:50:20

 Notes: XSELECT set up for      NICER
 Time keyword is TIME       in units of s
 Default timing binsize =   16.000

Setting…
 Image  keywords   = RAWX       RAWY        with binning =    1
 Energy keyword   = PI                      with binning =    1

Getting Min and Max for Energy Column…
Got min and max for PI:     0   1500

Got the minimum time resolution of the read data:  0.40000E-07
MJDREF =  5.6658000777593E+04 with TIMESYS = TDB
 Number of files read in:            1

******************** Observation Catalogue ********************

Data Directory is: /home/cananden/nicer-
data/1RXS_J154439.4-112820/1030120110/xti/event_cl/
HK Directory is: /home/cananden/nicer-
data/1RXS_J154439.4-112820/1030120110/xti/event_cl/

nicer:NICER-XTI-PHOTON > extract curve
extractor v6.11    12 Dec 2023
 Getting FITS WCS Keywords
 Doing file: /home/cananden/nicer-data/1RXS_J154439.4-
112820/1030120110/xti/event_cl/ni1030120110_0mpu7_cl_bary.evt
100% completed
          Total      Good    Bad: Time     Phase     Grade       Cut
           4702      4702            0         0         0         0
================================================================================
    Grand Total      Good    Bad: Time     Phase     Grade       Cut
           4702      4702            0         0         0         0
   in  2147.2     seconds
 Fits light curve has   4702.     counts for   2.190     counts/sec
 Thresholding removed more than half the bins
 Try exposure=0.0 on the extract command in xselect
 or lcthresh=0.0 if running extractor stand-alone
```

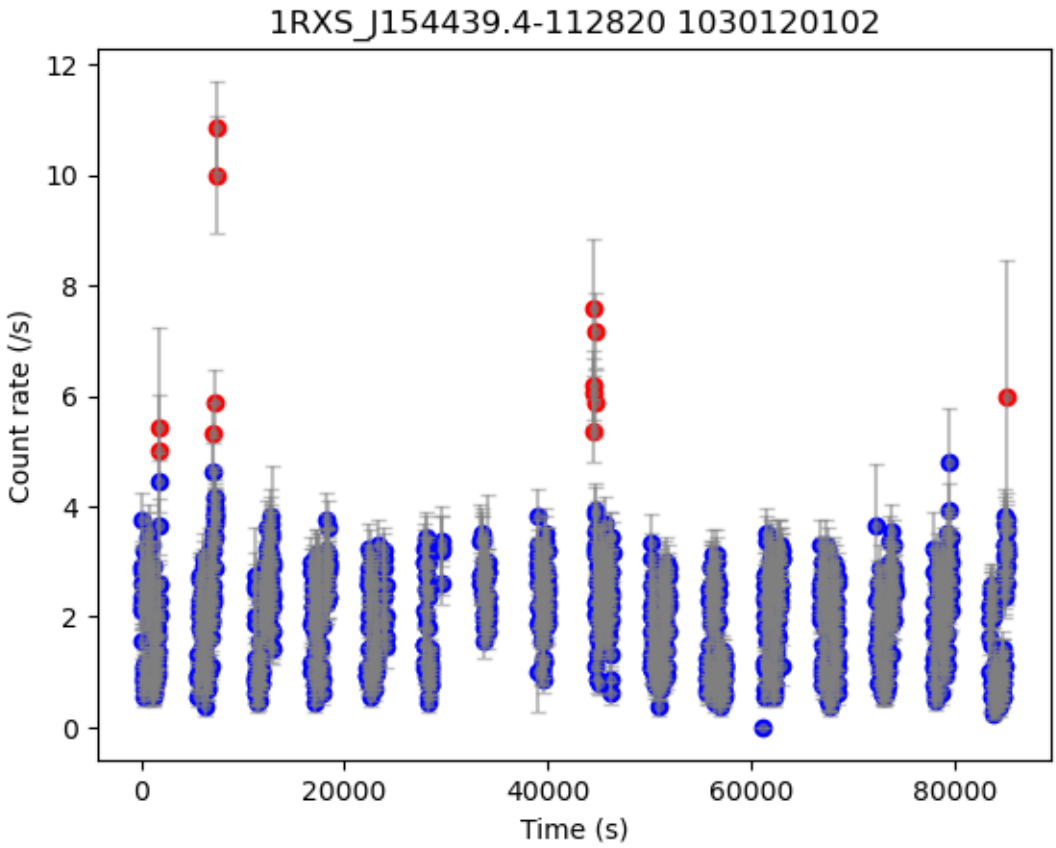
```
nicer:NICER-XTI-PHOTON > extract curve exposure=0.0
extractor v6.11    12 Dec 2023
 Getting FITS WCS Keywords
 Doing file: /home/cananden/nicer-data/1RXS_J154439.4-
112820/1030120110/xti/event_cl/ni1030120110_0mpu7_cl_bary.evt
100% completed
          Total      Good    Bad: Time     Phase     Grade       Cut
           4702      4702            0         0         0         0
================================================================================
    Grand Total      Good    Bad: Time     Phase     Grade       Cut
           4702      4702            0         0         0         0
   in  2147.2     seconds
 Fits light curve has   4702.     counts for   2.190     counts/sec
 Thresholding removed more than half the bins
 Try exposure=0.0 on the extract command in xselect
 or lcthresh=0.0 if running extractor stand-alone
nicer:NICER-XTI-PHOTON > save curve 1030120110_bary.fits
Wrote FITS light curve to file 1030120110_bary.fits
nicer:NICER-XTI-PHOTON > exit
> Save this session? >[n]
```

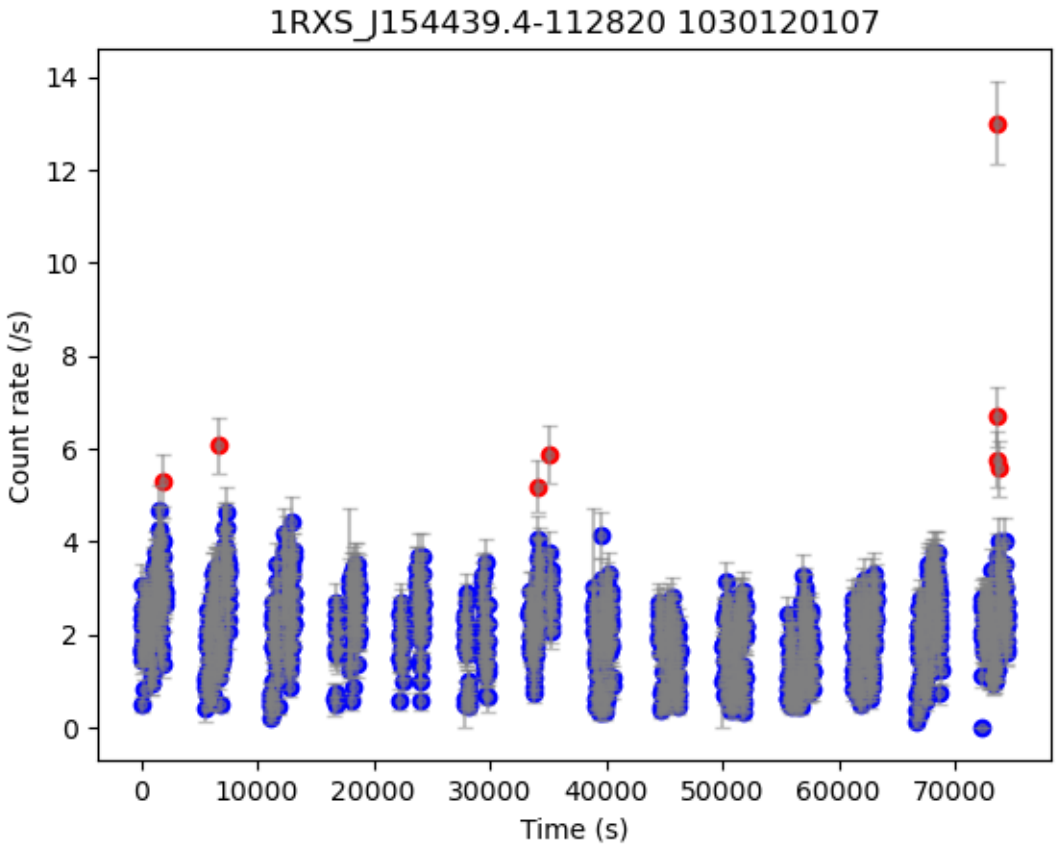
[12]:
```python
medians = []
errors = []
std = []

for i in range(len(ObsID_array)):
    os.chdir(f"/home/cananden/nicer-data/{star_folder}/{ObsID_array[i]}/xti")
    plot = fits.open(f"{ObsID_array[i]}.fits")
    data = plot[1].data

    temp_median = np.median(data["RATE"])
    temp_error  = np.median(data["ERROR"])

    medians = np.append(medians, temp_median)
    errors  = np.append(errors, temp_error)

    std_temp = np.std(data["RATE"])
    std = np.append(std, std_temp)
    continue

print(f"Medians: {medians}")
print(f"Errors: {errors}")
print(f"Standard Deviation: {std}")
```

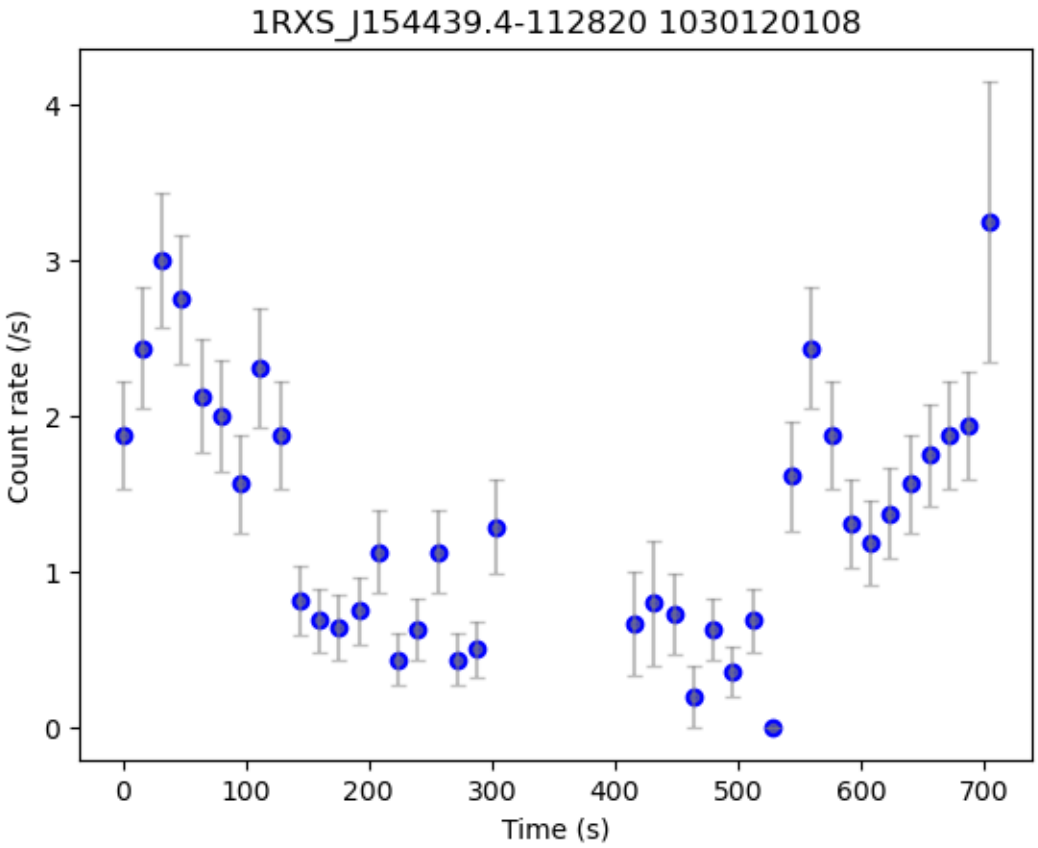
```
Medians: [2.83482146 2.         2.         1.28571427 1.875     ]
Errors: [0.42618704 0.35903516 0.35903516 0.30304575 0.34798527]
Standard Deviation: [0.97232926 0.96741712 0.95449716 0.80484664 0.63328445]
```

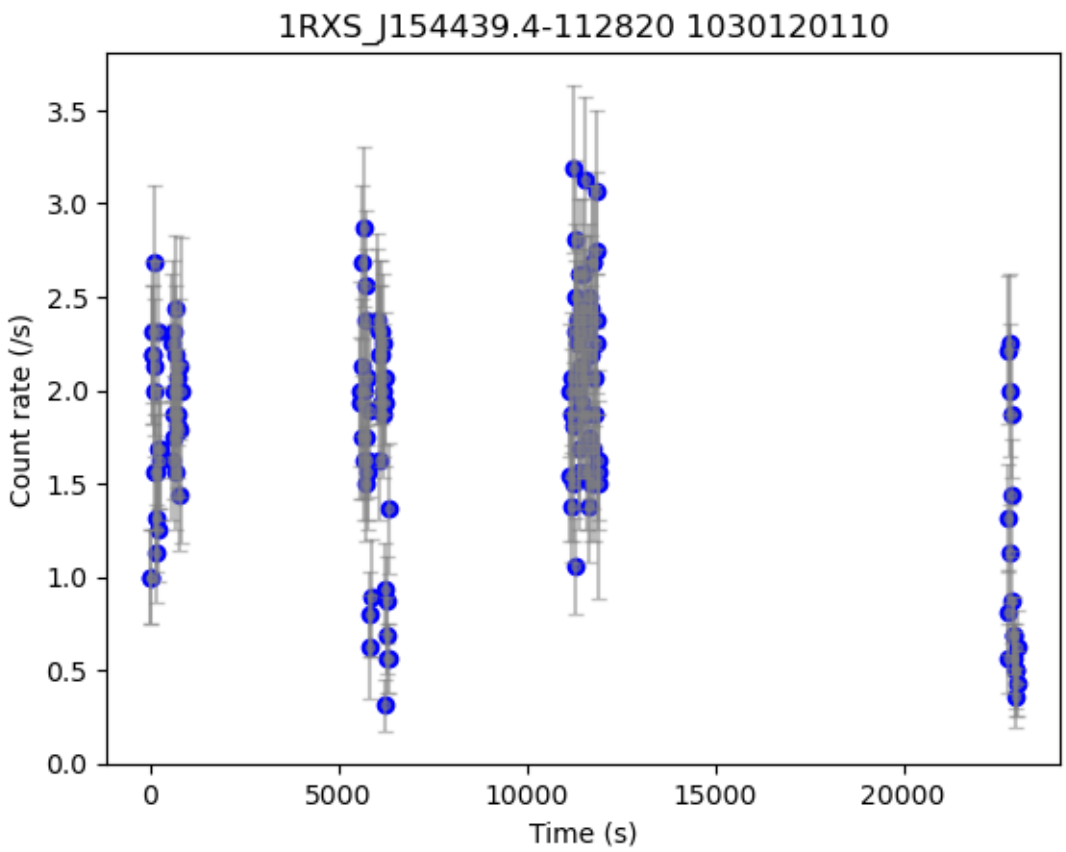
[13]:
```
os.chdir("/home/cananden/nicer-data/")
os.chdir(f"{star_folder}")

for i in range(len(ObsID_array)):
    os.chdir(f"{ObsID_array[i]}")
    os.chdir("xti")

    plot = fits.open(f"{ObsID_array[i]}.fits")
    data = plot[1].data

    three_sigma = 3 * std[i]
    medians_plot = medians[i]

    flares_filter = (data["RATE"] - medians_plot) > three_sigma
    mistakes = (medians_plot - data["RATE"]) > three_sigma
    normal = ~(flares_filter | mistakes)

    plt.scatter(data["TIME"][flares_filter], data["RATE"][flares_filter],␣
 ↪color="red")
    plt.scatter(data["TIME"][mistakes], data["RATE"][mistakes], color="gray")
    plt.scatter(data["TIME"][normal], data["RATE"][normal], color="blue")

    plt.xlabel("Time (s)")
    plt.ylabel("Count rate (/s)")

    plt.errorbar(data["TIME"], data["RATE"], yerr=data["ERROR"], fmt=".",␣
 ↪color="gray",
                 capsize=3, capthick=1, alpha=0.5)

    plt.title(f"{star_name} {ObsID_array[i]}")
    plt.show()
    plot.close()
    os.chdir("..")
    os.chdir("..")
```


1RXS_J154439.4-112820 0030120107
Count rate (/s)
Time (s)
0
1
2
3
4
5
0
5000
10000
15000
20000


1RXS_J154439.4-112820 1030120102
Count rate (/s)
Time (s)
0
2
4
6
8
10
12
0
20000
40000
60000
80000


1RXS_J154439.4-112820 1030120107
Count rate (/s)
Time (s)
0
2
4
6
8
10
12
14
0
10000
20000
30000
40000
50000
60000
70000


1RXS_J154439.4-112820 1030120108
Count rate (/s)
Time (s)
0
1
2
3
4
0
100
200
300
400
500
600
700



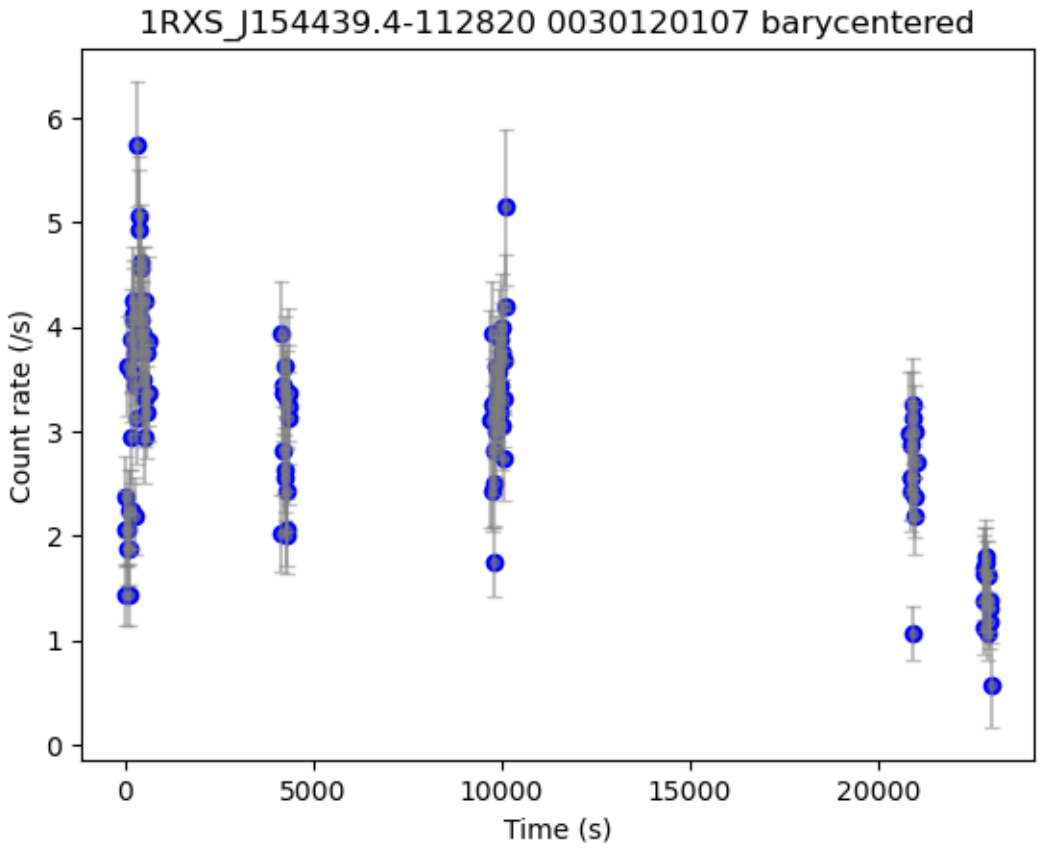
[14]:
```
medians_bary = []
errors_bary = []
std_bary = []

for i in range(len(ObsID_array)):
    os.chdir(f"/home/cananden/nicer-data/{star_folder}/{ObsID_array[i]}/xti")
    plot = fits.open(f"{ObsID_array[i]}_bary.fits")
    data = plot[1].data

    temp_median = np.median(data["RATE"])
    temp_error  = np.median(data["ERROR"])

    medians_bary = np.append(medians_bary, temp_median)
    errors_bary  = np.append(errors_bary, temp_error)

    std_temp = np.std(data["RATE"])
    std_bary = np.append(std_bary, std_temp)
    continue

print(f"Medians: {medians_bary}")
```

```
print(f"Errors: {errors_bary}")
print(f"Standard Deviation: {std_bary}")
```

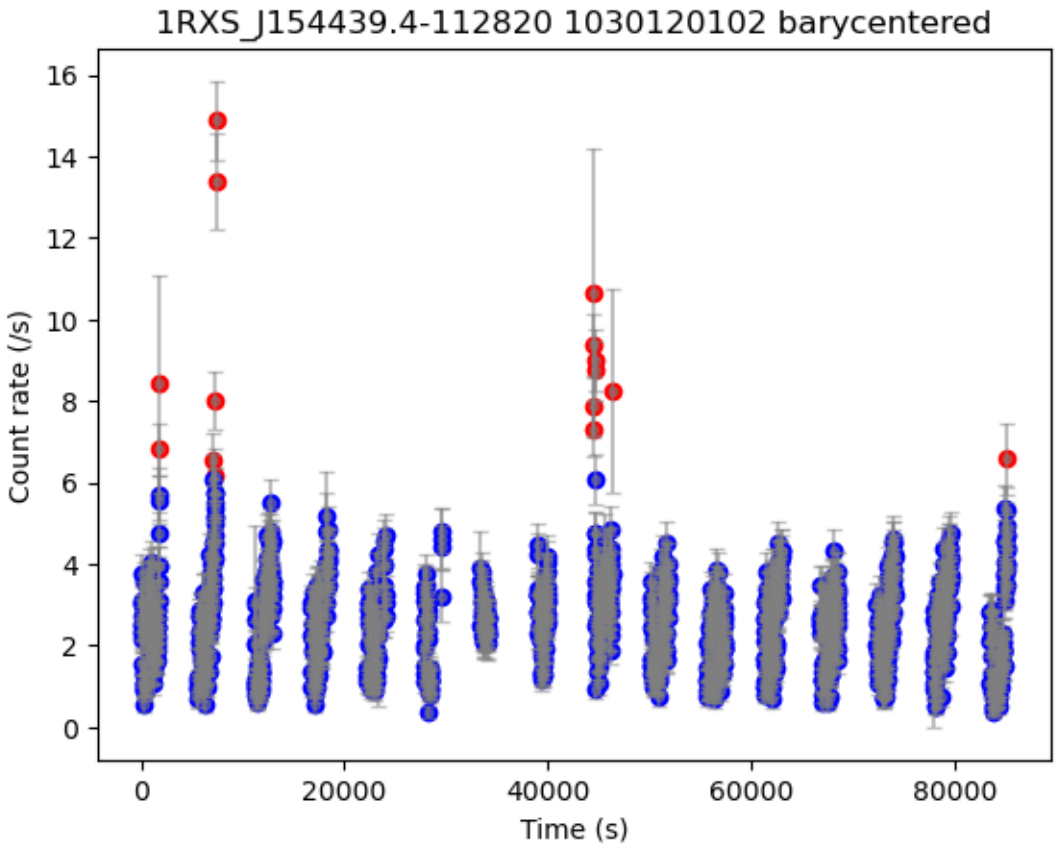
```
Medians: [3.125  2.5    2.4375 1.6875 2.1875]
Errors: [0.44633928 0.39528471 0.39031237 0.35355338 0.375     ]
Standard Deviation: [1.02302396 1.22222316 1.11558867 0.81596899 0.68080264]
```

[15]:
```
os.chdir("/home/cananden/nicer-data/")
os.chdir(f"{star_folder}")

for i in range(len(ObsID_array)):
    os.chdir(f"{ObsID_array[i]}")
    os.chdir("xti")

    plot = fits.open(f"{ObsID_array[i]}_bary.fits")
    data = plot[1].data

    print(np.max(data["TIME"]))

    three_sigma = 3 * std_bary[i]
    medians_plot = medians_bary[i]

    flares_filter = (data["RATE"] - medians_plot) > three_sigma
    mistakes = (medians_plot - data["RATE"]) > three_sigma
    normal = ~(flares_filter | mistakes)

    plt.scatter(data["TIME"][flares_filter], data["RATE"][flares_filter],␣
 ↪color="red")
    plt.scatter(data["TIME"][mistakes], data["RATE"][mistakes], color="gray")
    plt.scatter(data["TIME"][normal], data["RATE"][normal], color="blue")

    plt.xlabel("Time (s)")
    plt.ylabel("Count rate (/s)")

    plt.errorbar(data["TIME"], data["RATE"], yerr=data["ERROR"], fmt=".",␣
 ↪color="gray",
                 capsize=3, capthick=1, alpha=0.5)

    plt.title(f"{star_name} {ObsID_array[i]} barycentered")
    plt.show()
    plot.close()
    os.chdir("..")
    os.chdir("..")
```

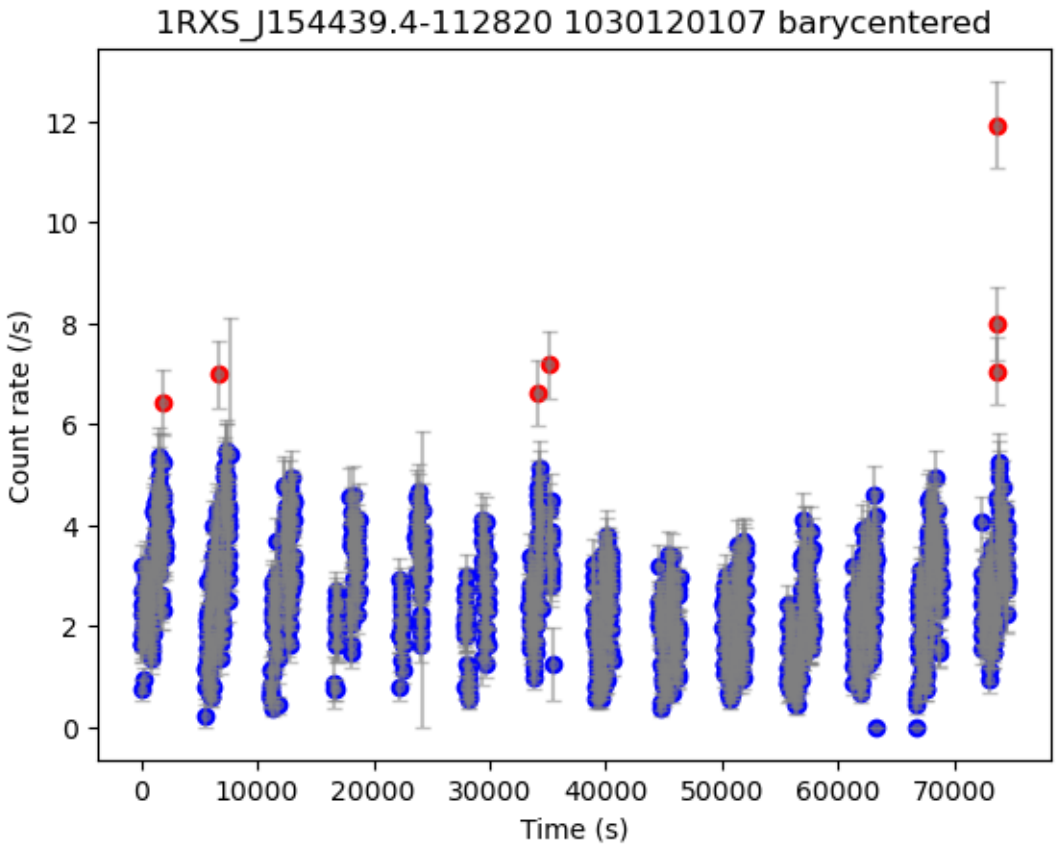
```
22976.0
```


1RXS_J154439.4-112820 0030120107 barycentered
Count rate (/s)
Time (s)
0
1
2
3
4
5
6
0
5000
10000
15000
20000

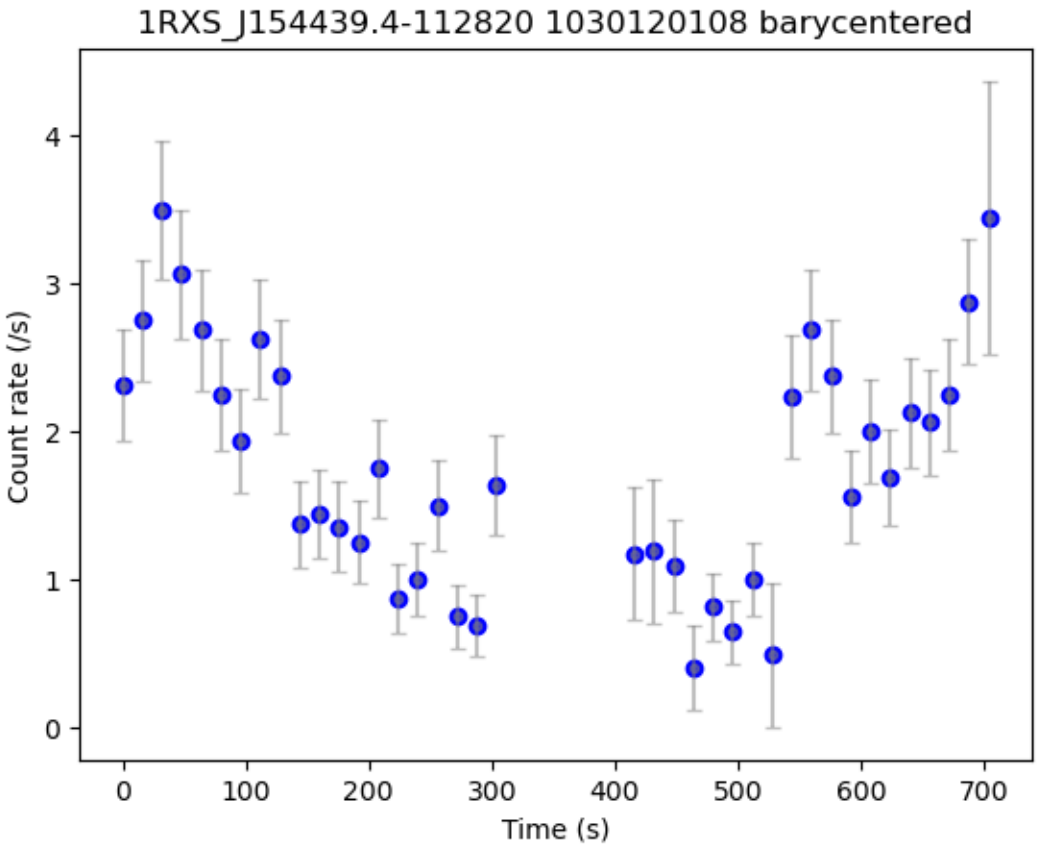
```
85184.0
```


1RXS_J154439.4-112820 1030120102 barycentered
Count rate (/s)
Time (s)
0
2
4
6
8
10
12
14
16
0
20000
40000
60000
80000

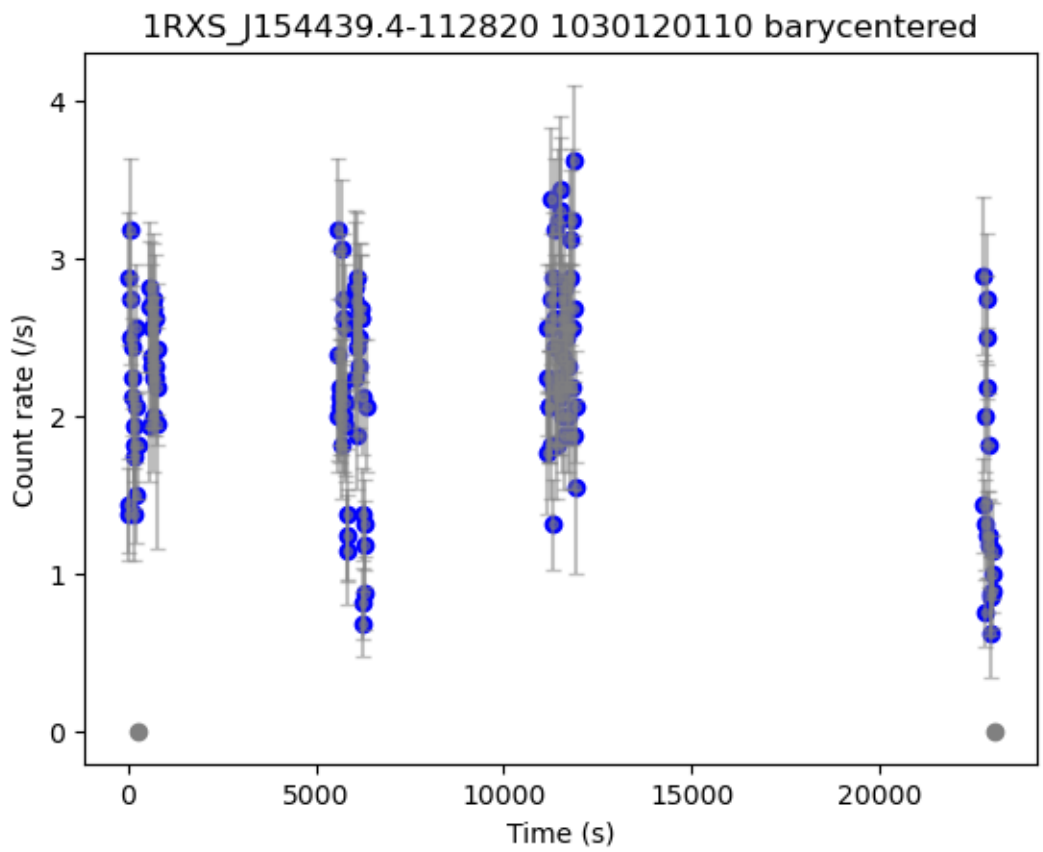
```
74560.0
```


1RXS_J154439.4-112820 1030120107 barycentered
Count rate (/s)
Time (s)
0
2
4
6
8
10
12
0
10000
20000
30000
40000
50000
60000
70000

```
704.0
```


1RXS_J154439.4-112820 1030120108 barycentered
Count rate (/s)
Time (s)
0
1
2
3
4
0
100
200
300
400
500
600
700

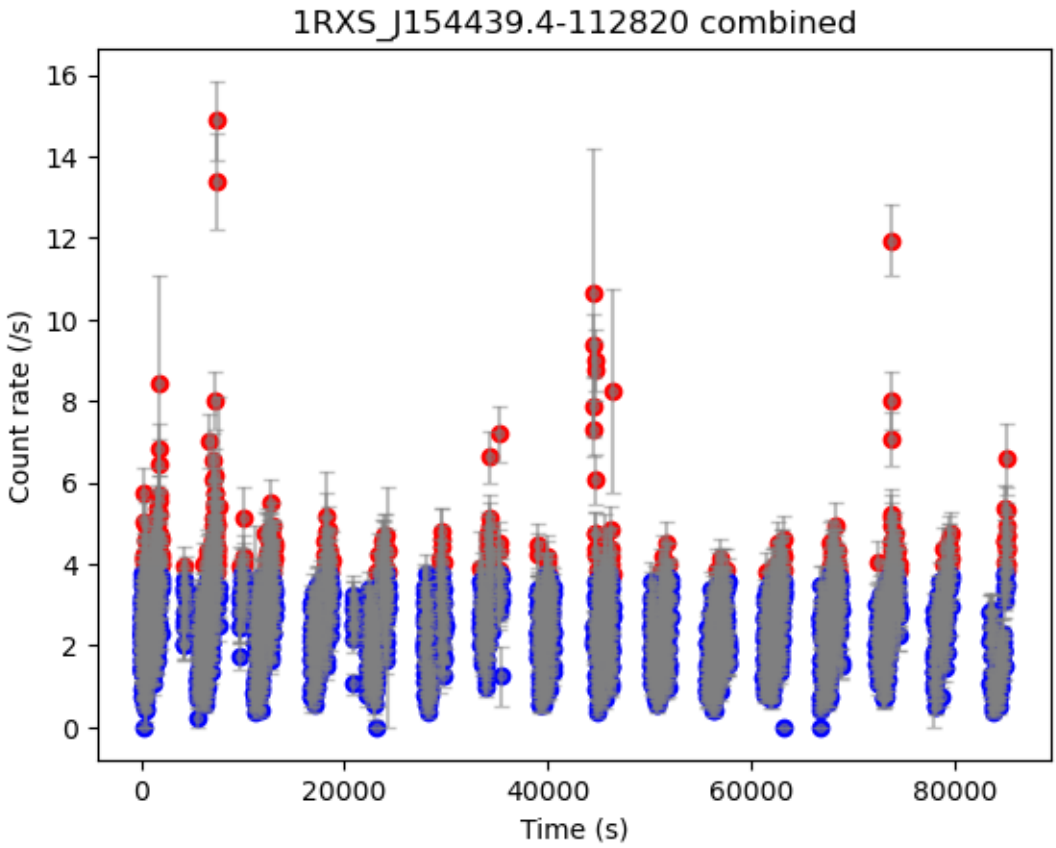
23040.0



[112]:
```
os.chdir("/home/cananden/nicer-data/")
os.chdir(f"{star_folder}")

if os.path.isfile(f"combined.fits"):
    print("Already combined")


else:

    file_array = []

    for i in range(len(ObsID_array)):
        next_file = (
        f"/home/cananden/nicer-data/{star_folder}/"
        f"{ObsID_array[i]}/xti/{ObsID_array[i]}_bary.fits"
    )

        file_array = np.append(file_array, next_file)
```

```
    tables = []

    for f in file_array:

        hdul = fits.open(f)
        tables.append(hdul[1].data)

    tables = [np.array(t) for t in tables]
    merged = np.hstack(tables)



    merged = np.sort(merged, order="TIME")

    hdu = fits.BinTableHDU(merged)
    hdu.writeto("combined.fits")
```

[113]:
```
os.chdir("/home/cananden/nicer-data/")
os.chdir(f"{star_folder}")

plot = fits.open(f"combined.fits")
data = plot[1].data

three_sigma = 3*std[i]
medians_plot = medians[i]
flares_filter = (data["RATE"] - medians_plot) > three_sigma
mistakes = (medians_plot - data["RATE"]) > three_sigma
normal = ~(flares_filter | mistakes)

plt.scatter(data["TIME"][flares_filter], data["RATE"][flares_filter], color =␣
 ↪"red")
plt.scatter(data["TIME"][mistakes], data["RATE"][mistakes], color = "gray")
plt.scatter(data["TIME"][normal], data["RATE"][normal], color = "blue")
plt.xlabel("Time (s)")
plt.ylabel("Count rate (/s)")
plt.errorbar(data["TIME"], data["RATE"], yerr=data["ERROR"], fmt=".",␣
 ↪color="gray", capsize=3, capthick=1, alpha = 0.5)
plt.title(f"{star_name} combined")

plt.show()
os.chdir("..")
os.chdir("..")
```



[119]:
```
colname = ObsID_complete.colnames[0]

Epoch_list = []
combined_epoch = pd.DataFrame()

week = timedelta(weeks=1)

for value in ObsID_array:
    row = ObsID_complete[ObsID_complete[colname] == value][0]
    Epoch_list.append(datetime.fromisoformat(str(row["Epoch"])))
print(Epoch_list)

current_group = [Epoch_list[0]]


for i in range(1, len(Epoch_list)):
    if Epoch_list[i] - current_group[0] <= week:
        current_group.append(Epoch_list[i])
    else:
```

```
        combined_epoch = pd.concat([combined_epoch, pd.DataFrame({"group":␣
↪[current_group]})], ignore_index = True)



        current_group = [Epoch_list[i]]

combined_epoch = pd.concat([combined_epoch, pd.DataFrame({"group":␣
↪[current_group]})], ignore_index = True)

print(combined_epoch.iloc[1])

print(len(combined_epoch))
```

```
[datetime.datetime(2017, 7, 1, 7, 7, 30), datetime.datetime(2018, 1, 25, 23,
43), datetime.datetime(2018, 1, 31, 0, 11, 24), datetime.datetime(2018, 2, 6,
21, 58, 20), datetime.datetime(2018, 2, 9, 14, 50, 20)]
group    [2018-01-25 23:43:00, 2018-01-31 00:11:24]
Name: 1, dtype: object
3
```

[133]:
```
os.chdir("/home/cananden/nicer-data/")
os.chdir(f"{star_folder}")

epoch_to_obsid = {Epoch_list[k]: ObsID_array[k] for k in range(len(Epoch_list))}

for i in range(len(combined_epoch)):


    outname = f"combined_{i}.fits"

    if os.path.isfile(f"{outname}"):
        print(f"Group {i} Already combined")


    else:
        file_array = []

        for j in range(len(combined_epoch.iloc[i].iloc[0])):

            epoch = combined_epoch.iloc[i].iloc[0][j]
            obsid = epoch_to_obsid[epoch]

            next_file = (
                f"/home/cananden/nicer-data/{star_folder}/"
                f"{obsid}/xti/{obsid}_bary.fits"
            )
```

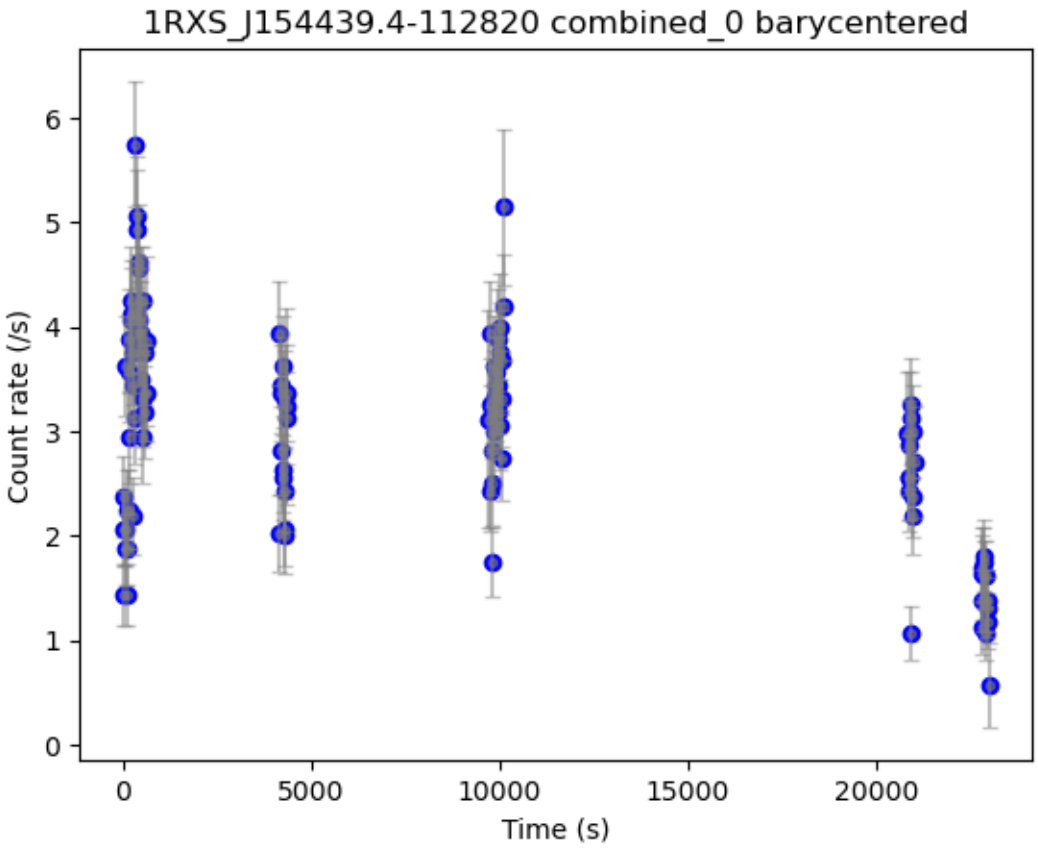
```
            file_array = np.append(file_array, next_file)

        tables = []

        for f in file_array:
            hdul = fits.open(f)
            tables.append(hdul[1].data)

        tables = [np.array(t) for t in tables]
        merged = np.hstack(tables)
        merged = np.sort(merged, order="TIME")

        hdu = fits.BinTableHDU(merged)
        hdu.writeto(outname)
```

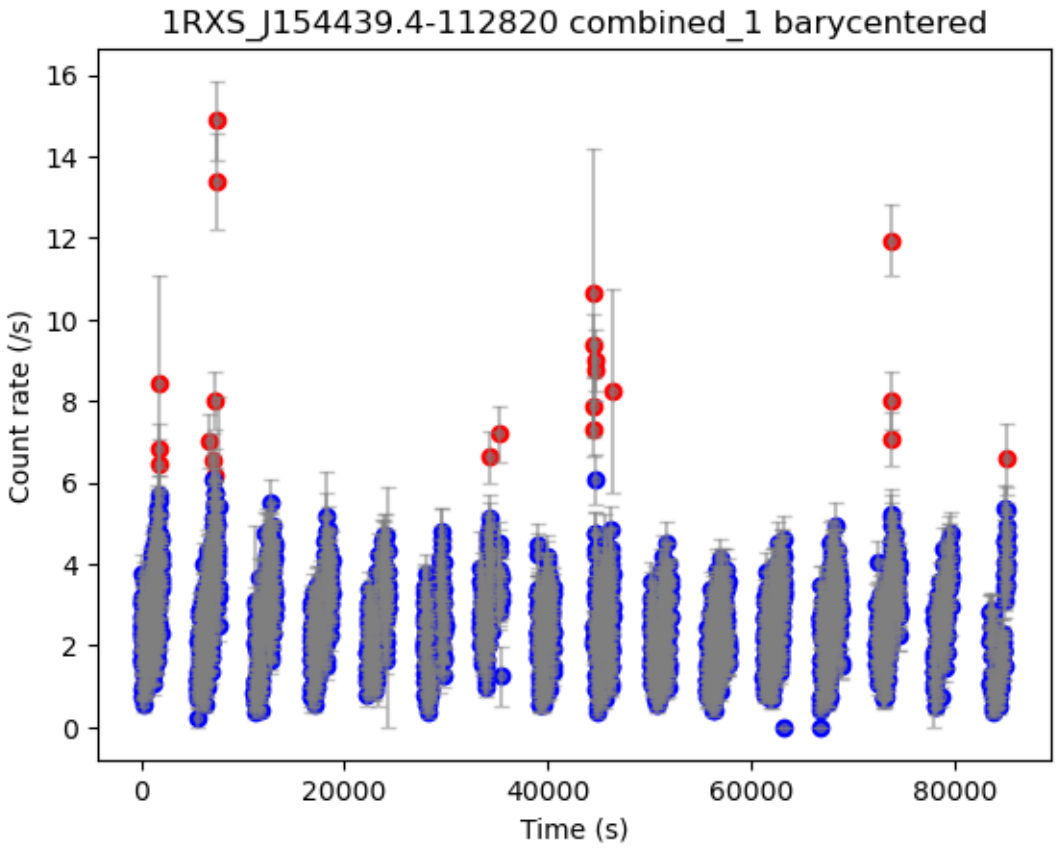
```
Group 0 Already combined
Group 1 Already combined
Group 2 Already combined
```

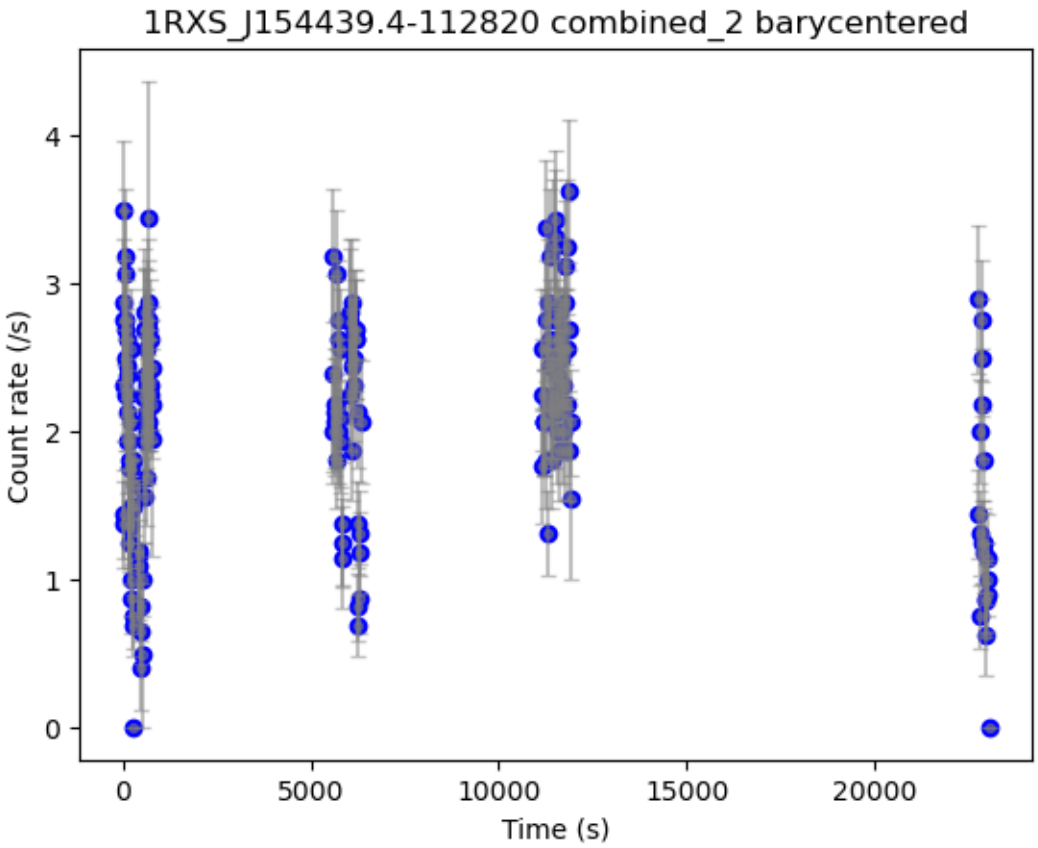
[139]:
```
medians_combined = []
errors_combined = []
std_combined = []

for i in range(len(combined_epoch)):

    outname = f"combined_{i}.fits"
    path = f"/home/cananden/nicer-data/{star_folder}/{outname}"

    plot = fits.open(path)
    data = plot[1].data

    temp_median = np.median(data["RATE"])
    temp_error  = np.median(data["ERROR"])
    temp_std    = np.std(data["RATE"])

    medians_combined = np.append(medians_combined, temp_median)
    errors_combined  = np.append(errors_combined, temp_error)
    std_combined     = np.append(std_combined, temp_std)

print(f"Medians: {medians_bary}")
print(f"Errors: {errors_bary}")
print(f"Standard Deviation: {std_bary}")
```

```
Medians: [3.125  2.5    2.4375 1.6875 2.1875]
Errors: [0.44633928 0.39528471 0.39031237 0.35355338 0.375     ]
Standard Deviation: [1.02302396 1.22222316 1.11558867 0.81596899 0.68080264]
```

[140]:
```
os.chdir("/home/cananden/nicer-data/")
os.chdir(f"{star_folder}")

for i in range(len(combined_epoch)):


    plot = fits.open(f"combined_{i}.fits")
    data = plot[1].data



    three_sigma = 3 * std_bary[i]
    medians_plot = medians_bary[i]

    flares_filter = (data["RATE"] - medians_plot) > three_sigma
    mistakes = (medians_plot - data["RATE"]) > three_sigma
    normal = ~(flares_filter | mistakes)

    plt.scatter(data["TIME"][flares_filter], data["RATE"][flares_filter],␣
 ↪color="red")
    plt.scatter(data["TIME"][mistakes], data["RATE"][mistakes], color="gray")
    plt.scatter(data["TIME"][normal], data["RATE"][normal], color="blue")

    plt.xlabel("Time (s)")
    plt.ylabel("Count rate (/s)")

    plt.errorbar(data["TIME"], data["RATE"], yerr=data["ERROR"], fmt=".",␣
 ↪color="gray",
                 capsize=3, capthick=1, alpha=0.5)

    plt.title(f"{star_name} combined_{i} barycentered")
    plt.show()
    plot.close()
```


1RXS_J154439.4-112820 combined_0 barycentered
Count rate (/s)
Time (s)
0
1
2
3
4
5
6
0
5000
10000
15000
20000


1RXS_J154439.4-112820 combined_1 barycentered
Count rate (/s)
Time (s)
0
2
4
6
8
10
12
14
16
0
20000
40000
60000
80000


1RXS_J154439.4-112820 combined_2 barycentered
Count rate (/s)
Time (s)
0
1
2
3
4
0
5000
10000
15000
20000